

Structure-Adaptive Transfer Learning for Sparse Estimation

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Abstract

We study the problem of high-dimensional compound estimation under a transfer learning paradigm. Traditional approaches typically rely on the restrictive assumption of *value-similarity* between source and target parameters. In practice, however, parameters often exhibit significant differences in magnitude while sharing underlying *structural similarities*, such as overlapping sparsity patterns. To bridge this gap, we introduce a Structure-Adaptive Transfer Learning (SATL) framework, which extends the false discovery rate (FDR) thresholding methodology of [Abramovich et al. \(2006\)](#) to incorporate structural information via p-value weighting. SATL adaptively assigns coordinate-specific thresholds, using a data-driven weighting scheme that leverages external datasets to narrow down candidate signal regions while controlling false discoveries. Theoretically, we establish robustness guarantees under mild conditions, ensuring that SATL does not substantially underperform conventional estimators even with uninformative external data. When source data provide useful structural information, SATL provably achieves faster convergence rates than standard estimators that ignore auxiliary information, and delivers strong performance even when the external data are only partially informative. Our numerical experiments show that SATL achieves substantial improvements in estimation accuracy and reliably enables transfer learning in scenarios where conventional value-based methods fail.

Key words and phrases: Adaptive Shrinkage Estimation; False Discovery Rate; Negative Transfer; Structural Similarity; Weighted p-value.

1 Introduction

1.1 Sparse Estimation and Soft-Thresholding

We study the problem of compound estimation of normal means in high-dimensional settings, a classical statistical topic with wide-ranging applications. Foundational contributions to this area include the development of linear shrinkage estimators (James and Stein, 1961; Efron and Morris, 1975), thresholding-based methods (Donoho and Johnstone, 1994; Abramovich et al., 2006; Cai and Zhou, 2009), and empirical Bayes approaches (Brown and Greenshtein, 2009; Efron, 2011; Koenker and Mizera, 2014). These methodologies not only offer deep insights into fundamental estimation challenges but also have significant implications for other statistical learning tasks such as regression, classification and clustering.

A key characteristic of modern high-dimensional data is *sparsity*, where only a small fraction of study units have meaningfully large effects, while the majority have negligible or zero effects. This structural property arises naturally across diverse domains: in astronomical image analysis, pixel arrays of celestial objects frequently contain extensive dark regions, with only limited areas displaying non-zero intensities; in genomic research, among thousands of measured genes, typically only a small subset demonstrate significant expression levels associated with specific biological processes; in signal processing, although a wide array of candidate predictors may be available, the underlying signal model generally depends on a sparse subset of relevant components.

Sparsity can be formally characterized in several ways. The most intuitive notion considers the proportion of non-zero elements. Let k_n denote the number of non-zero signals and n the total dimension. The ratio k_n/n serves as a sparsity index, with smaller values indicating stronger sparsity levels. This paper primarily focuses on this definition of sparsity,

although alternative characterizations, such as through weak or strong ℓ_p balls, also exist in the literature (cf. [Abramovich et al., 2006](#)).

Thresholding methods have emerged as particularly effective tools in high-dimensional inference ([Donoho and Johnstone, 1994](#); [Johnstone and Silverman, 2004](#); [Bickel and Levina, 2008](#); [Cai and Liu, 2011](#); [Fan et al., 2011](#)), offering both accurate estimation and enhanced interpretability by systematically suppressing noise while preserving significant signals. The soft-thresholding operator is a canonical example of this paradigm. Let Y denote a generic variable with an expected value μ , and σ be the standard deviation. Define the thresholding function $\delta(y, t)$ as follows:

$$\delta(Y, t) = \begin{cases} -Y\sigma^{-1}, & \text{if } |Y\sigma^{-1}| \leq t, \\ -t \cdot \text{sign}(Y\sigma^{-1}), & \text{otherwise.} \end{cases} \quad (1.1)$$

The corresponding soft-thresholding estimator of μ takes the form $\hat{\mu}(Y, t) = Y + \sigma\delta(Y, t)$. Unless otherwise stated, the variance in (1.1) is fixed to 1. This construction forms the foundation for our proposed method.

1.2 Transfer Learning for Sparse Estimation: Background and Motivation

Transfer learning is a highly effective approach for improving inference on target domains by utilizing information from related source domains ([Pan and Yang, 2010](#); [Torrey and Shavlik, 2010](#); [Zhuang et al., 2021](#)). It has demonstrated considerable promise across a wide range of applications, such as genomics, medical imaging, and document classification. Let $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)$ denote the target parameter vector, and $\boldsymbol{\mu}^{(m)} = (\mu_1^{(m)}, \dots, \mu_n^{(m)})$ represent the parameters from the m -th external dataset, $m \in [M]$. Most existing transfer learning techniques depend heavily on the assumption of *value-similarity*, positing that the target parameter $\boldsymbol{\mu}$ can be closely approximated by (one or) some combinations of source

parameters $\{\boldsymbol{\mu}^{(m)}\}_{m=1}^M$ from M different related domains. However, this assumption can be overly restrictive in real-world contexts.

For instance, in RNA-seq analysis, gene expression profiles from target and source studies, such as data collected from heart disease patients (as the target domain) and from diabetes patients (as the source domain), may exhibit substantial overlap in the sets of actively expressed genes related to both diseases. However, the actual expression levels can differ markedly due to various factors, including batch effects, clinical heterogeneity, or technical discrepancies across sequencing platforms. Consequently, the ℓ_1 distance $\|\boldsymbol{\mu} - \boldsymbol{\mu}^{(m)}\|_1$ may remain large. Such a violation of the value-similarity condition not only compromises the effectiveness of transfer learning methods but may also lead to *negative transfer*, wherein incorporating source data degrades performance in the target task.

In contrast, our work introduces a weaker and more flexible notion of *structural similarity*. This concept requires that the source and target parameters share certain structural traits, such as sparsity patterns or support sets, without imposing constraints on their numerical values. This structural transfer paradigm allows for significant differences in parameter magnitudes and applies to a wide variety of applications. Two exemplar applications include:

- (a) *Image Processing*: Objects that appear “nearly black” under different lighting conditions may have very different pixel intensities but retain similar non-zero pixel locations.
- (b) *Genomics*: In genetic studies of cardiovascular disease (CVD), genes implicated in type 2 diabetes may play a role in CVD pathogenesis. Though gene expression levels vary across patient groups, active or disease-associated genomic regions often show substantial overlap for different diseases.

Although transfer learning has progressed significantly in high-dimensional regression, graphical models, and nonparametric estimation (Li et al., 2021; Hu and Zhang, 2023; Li et al., 2023; Cai and Pu, 2024), existing methods primarily rely on the notion of value-similarity and are not directly applicable to compound estimation problems where sparsity

patterns are the primary form of shared information. This motivates a new line of inquiry: How can we systematically adapt existing thresholding techniques for sparse estimation to incorporate external knowledge through transfer learning frameworks, particularly when source and target tasks share only underlying structural similarities rather than precise value similarities?

1.3 Structure-Adaptive Transfer Learning

Classical thresholding procedures (Johnstone and Silverman, 2004; Johnstone, 2011) typically apply a universal threshold across all coordinates. In real-world transfer learning settings, however, sparsity patterns often exhibit considerable heterogeneity. This motivates our proposal of the structure-adaptive transfer learning (SATL) framework, which extends conventional soft-thresholding methodologies by incorporating structural information from source domains to set data-adaptive, heterogeneous thresholds.

Our approach builds upon the innovative sparsity-adaptive thresholding framework of Abramovich et al. (2006), which bridges decision-theoretic sparse estimation with false discovery rate (FDR) methodology (Benjamini and Hochberg, 1995). The FDR paradigm provides a principled approach for determining data-driven thresholds and optimally adapts to unknown sparsity (Abramovich et al., 2006). Moreover, as we rigorously establish in subsequent sections, the FDR concept is particularly suitable for transfer learning settings by enabling the effective integration of structural information through p-value weighting.

The core innovation of SATL lies in leveraging structural similarities (i.e., overlaps in sparsity patterns) between target and source parameters through a principled framework of weighted FDR analysis to set varied thresholds. This approach allows us to incorporate shared structural information while accommodating substantial disparities in specific parameter values. The SATL procedure operates through two stages:

- *Structural Learning*: We estimate the support of each external parameter $\mu^{(m)}$ using a screening procedure. The estimated supports are aggregated to form a candidate

support set for the target parameter. From this set, we derive a weight vector that encodes the structural information extracted from the source domains.

- *Adaptive Thresholding*: The weights are integrated with the target data through a weighted FDR-thresholding algorithm. This algorithm adapts simultaneously to both the global sparsity of the target parameter and the local heterogeneity informed by external data.

Our work makes several contributions to the field of sparse estimation. First, we introduce a robust and principled SATL framework that incorporates a sparsity-adaptive FDR thresholding estimator, along with a practical approach for constructing a candidate support set through the use of external datasets. Second, we extend the minimax decision theory of [Johnstone \(2011\)](#), a cornerstone in sparse inference ([Abramovich et al., 2006](#); [Tibshirani, 2014](#); [Collier et al., 2017](#)), to derive novel higher-order characterizations of the maximal risk for soft-thresholding estimators within the transfer learning context. This enables us to identify specific conditions under which SATL provably outperforms traditional minimax estimators. Third, we demonstrate that when source data contain meaningful structural knowledge, SATL achieves faster convergence rates than standard estimators that do not incorporate auxiliary information. Fourth, our theoretical framework imposes only mild conditions to avoid negative transfer, demonstrating the robustness and broad applicability of SATL. In particular, SATL remains effective even in scenarios where conventional transfer learning methods based on value similarity fail. Finally, extensive numerical experiments show that our estimator significantly improves efficiency compared to existing methods.

1.4 Related Work

Our SATL framework brings together several key concepts from related directions, including FDR thresholding, covariate-assisted inference, and high-dimensional transfer learning. We briefly discuss the connections and distinctions with each of these research areas.

Sparse estimation with FDR thresholding. The seminal work of [Abramovich et al. \(2006\)](#) introduced an FDR-thresholding framework that adaptively estimates sparse signals and achieves the minimax optimal rate. This approach was later generalized to high-dimensional regression through the SLOPE procedure ([Bogdan et al., 2015](#)), which coincides with FDR thresholding under orthogonal designs and has been shown to outperform conventional regression methods in sparse settings ([Su and Candès, 2016](#); [Kos and Bogdan, 2020](#); [Hu and Lu, 2022](#)). Motivated by these advances, we extend the adaptive thresholding paradigm to a transfer learning setting, incorporating auxiliary external data to enhance the efficacy of sparse estimation through a novel structural transfer strategy.

Structured multiple testing. Our strategy for weight construction is grounded in principles from multiple testing and weighted FDR methodology, ensuring both theoretical rigor and practical effectiveness. Specifically, our approach is inspired by a fruitful line of research that leverages side information to improve the efficiency and interpretability of FDR analyses ([Genovese et al., 2006](#); [Roquain and Van De Wiel, 2009](#); [Cai et al., 2019](#); [Li and Barber, 2019](#); [Ignatiadis and Huber, 2021](#); [Cai et al., 2022](#)). We exploit tools related to weighted p-values from these works to set varied, data-adaptive thresholds that incorporate auxiliary data into soft-thresholding estimators. Particularly relevant to our setting is the ASUS method ([Banerjee et al., 2020](#)), which constructs auxiliary statistics to partition hypotheses into groups with varying sparsity levels and applies group-specific thresholds to adapt to local sparsity. However, ASUS is inherently limited to a single auxiliary sequence and cannot accommodate multiple auxiliary sources, a constraint that undermines its applicability to multi-source transfer learning settings such as those considered in this paper.

High-dimensional transfer learning. This area has attracted significant attention recently. Early work focused on single-source linear regression ([Bastani, 2021](#)), with subsequent extensions to a wide variety of models and multiple sources. Examples include high-dimensional linear models ([Li et al., 2021](#); [Lin et al., 2024](#); [Zhao et al., 2024](#)), general-

ized linear models (Tian and Feng, 2023; Li et al., 2024), semiparametric models (Hu and Zhang, 2023), nonparametric models (Lin and Reimherr, 2024; Cai and Pu, 2024; Auddy et al., 2025), quantile regression (Zhang and Zhu, 2025), panel regression (Duan et al., 2024), and spatial autoregression (Zeng et al., 2024), among others. These approaches primarily exploit algebraic relationships between target and source parameters to attain minimax rates under specific parameter spaces. In contrast, our framework focuses on transferring sparsity structures, offering a more flexible scheme that does not require any functional form linking μ to the external parameters.

1.5 Organization of the Paper

The article is organized as follows. Section 2 describes the basic framework. Section 3 presents data-driven SATL algorithm and its theoretical analyses. The numerical performance of SATL is investigated using simulated data and an RNA-seq dataset in Sections 4 and 5, respectively. Additional numerical results and proofs are given in the Supplementary Materials.

2 Structural Transfer Learning: The Basic Framework

This section first describes the model and assumptions, then discusses a general strategy for structural transfer based on weighted thresholding, and finally discusses principles for weight construction and establishes conditions under which structural transfer is robust against weight misspecification.

2.1 Warm-Up and Preliminaries

We investigate the sparse estimation problem under transfer learning framework. The goal is to estimate a high-dimensional mean $\mu \in \mathbb{R}^n$, given access to M external datasets with unknown means $\{\mu^{(m)}\}_{m=1}^M$. The summary statistics of target and external datasets

obey

$$\begin{aligned} y_i &= \mu_i + \sigma_i \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \\ y_i^{(m)} &= \mu_i^{(m)} + \sigma_i^{(m)} \epsilon_i^{(m)}, \quad \epsilon_i^{(m)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \end{aligned} \tag{2.2}$$

Without loss of generality, we set the variances for all coordinates to be one. Let $S = \{i \in [n] : \mu_i \neq 0\}$ and $[n] = \{1, 2, \dots, n\}$, and assume $k_n := |S|$ is much smaller than n . Denote the estimator by $\hat{\boldsymbol{\mu}} = (\hat{\mu}_1, \dots, \hat{\mu}_n)$. The squared error loss is given by $\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_2^2$. [Extensions to the unknown variance case and the non-Gaussian noise case are investigated in Section S6.](#)

We briefly outline the FDR-thresholding framework in [Abramovich et al. \(2006\)](#). Consider the simultaneous testing of n hypotheses: $H_{0,i} : \mu_i = 0$ vs. $H_{1,i} : \mu_i \neq 0$, $i \in [n]$. Let $R \subseteq [n]$ be the index set of rejections. The FDR is defined as

$$\text{FDR} = \mathbb{E} \left[\frac{\sum_{i=1}^n \mathbb{I}\{\mu_i = 0, i \in R\}}{|R| \vee 1} \right].$$

Let $p_{(1)} \leq \dots \leq p_{(n)}$ denote the ordered p-values. The Benjamini-Hochberg (BH) procedure ([Benjamini and Hochberg, 1995](#)) defines the threshold index for test level $q \in (0, 1)$ as

$$\ell = \max \left\{ j \in [n] : p_{(j)} \leq \frac{j}{n} q \right\}. \tag{2.3}$$

[Abramovich et al. \(2006\)](#) proposed a simple FDR-thresholding estimator that applies a uniform threshold to all units:

$$t_F = \tilde{\Phi}^{-1} \left(\frac{\ell q}{2n} \right), \tag{2.4}$$

where $\tilde{\Phi}^{-1}$ denotes the upper quantile function of the standard normal distribution. For example, substituting y_i and t_F into the soft-thresholding rule [\(1.1\)](#) yields $\hat{\mu}_i = y_i + \delta(y_i, t_F)$ for $i \in [n]$. The authors provide an elegant theoretical analysis, demonstrating that the class of FDR-based data-adaptive thresholding estimators is asymptotically minimax, adapting

to unknown sparsity across a spectrum of sparsity classes and risk functions.

However, the uniform threshold fails to accommodate local structural heterogeneity when auxiliary data are present. In transfer learning settings, the availability of external datasets necessitates data-adaptive and coordinate-specific thresholds to effectively capture structural similarities between source and target parameters. To this end, the next section discusses how to extend the FDR-thresholding framework of [Abramovich et al. \(2006\)](#) by incorporating ideas from structured multiple testing, thereby enabling the principled integration of side information into sparse estimation.

2.2 Weighted FDR Thresholding

To jointly utilize both the target and external data within the transfer learning framework, we extend the FDR-thresholding framework from [Abramovich et al. \(2006\)](#) to develop a weighted FDR thresholding strategy that sets varied thresholds across estimation units, motivated by methodologies in structured multiple testing ([Li and Barber, 2019](#); [Cai et al., 2022](#)). The proposed SATL framework consists of two steps: (a) learning structural knowledge about the target support through external datasets, and (b) transferring this knowledge by assigning coordinate-specific weights in the soft-thresholding estimator. We first describe a prototype procedure below, with instantiations and implementation details provided in [Section 3](#).

- (a) For each external dataset $\mathbf{y}^{(m)}$, apply a screening procedure to estimate the support set $S^{(m)}$, denoted as $\hat{S}^{(m)}$. Then obtain a candidate support set $\hat{S}$ by aggregating the estimators $\{\hat{S}^{(m)}\}_{m=1}^M$. Determine weights $\hat{w}_i$ for each coordinate based on $\hat{S}$.
- (b) Apply weighted FDR thresholding to the target data with coordinate-specific thresholds:

$$\hat{\mu}_i = \hat{\mu}(y_i, t(\hat{w}_i, \ell, q)) = y_i + \delta(y_i, t(\hat{w}_i, \ell, q)), \quad i \in [n], \quad (2.5)$$

where ℓ is the threshold index defined by [\(2.3\)](#), $t(\hat{w}_i, \ell, q)$ is a weighted FDR threshold

for coordinate i :

$$t(\hat{w}_i, \ell, q) = \tilde{\Phi}^{-1} \left\{ \frac{1}{2} \min \left(\hat{w}_i \frac{\ell q}{n}, 1 - \varepsilon \right) \right\}, \quad i \in [n], \quad (2.6)$$

for some small constant $\varepsilon \in (0, 1)$. The term $\min(1 - \varepsilon, \cdot)$ ensures that the threshold remains well-defined when $\hat{w}_i \ell q / n > 1$.

Now we provide some intuition behind the design of this prototype SATL procedure. In stage (a), we learn the signal-support structure from external datasets, which is then used to inform the weights of coordinate-wise thresholds. The data-driven method for aggregating information across multiple sources to estimate the support is detailed in Section 3. The key idea is that when the external parameters share structural similarity with the target parameter, the estimated support $\hat{S}$ serves as a reasonable proxy for the true support S . We then construct weights by up-weighting coordinates in $\hat{S}$ and down-weighting those in $\hat{S}^c$, which effectively lowers the thresholds for locations where signals are more probable.

Step (b) implements a weighted FDR-thresholding procedure. As in Abramovich et al. (2006), the FDR-based construction of the threshold preserves the advantage of adapting to the global sparsity level. The FDR level q determines the adaptive threshold $t(\hat{w}_i, \ell, q)$ and controls the extent of adaptivity inherited from the BH procedure (Abramovich et al., 2006): when signals are abundant, the data-dependent factor ℓ yields smaller thresholds, thereby improving adaptation to the overall sparsity structure. Compared with the universal threshold t_F in (2.4), which incorporates only the global factor $\ell q / (2n)$, the new threshold in (2.6) further integrates local structural information through the data-driven weights. This design is consistent with ideas in weighted multiple testing (Genovese et al., 2006; Roquain and Van De Wiel, 2009; Sun et al., 2015; Li and Barber, 2019; Cai et al., 2022), where hypotheses with stronger auxiliary evidence of being non-null are up-weighted to increase their likelihood of rejection. Correspondingly, in the thresholding framework, coordinates

that are more likely to contain signals receive lower thresholds. As a result, the estimator adapts simultaneously to the unknown global sparsity level k_n and to coordinate-specific sparsity patterns informed by auxiliary data.

2.3 Weighting Scheme and Robustness Theory

A central challenge in transfer learning is to effectively leverage auxiliary data while safeguarding against negative transfer. This section discusses guiding principles for implementing the SATL framework, focusing on *the design of weights* and *robustness guarantees*. Generally speaking, the weight assignment must be calibrated to appropriately prioritize units where auxiliary data provide credible structural evidence, ensuring that external information is meaningfully incorporated. Meanwhile, the estimation of the signal support must ensure that even if the estimated support deviates significantly from the true one, the resulting estimator will not substantially underperform standard methods that use no side information.

1. The design of weights. Given the estimated support $\hat{S}$, we assign larger weights to coordinates within $\hat{S}$ and smaller weights to those outside. The selection of weight magnitudes balances two objectives. On one hand, weights must be sufficiently large to meaningfully integrate structural information, effectively lowering thresholds for likely signals and improving estimation efficiency. On the other hand, they must be relatively conservative in magnitudes to safeguard against negative transfer in cases where $\hat{S}$ is inaccurately estimated. Our weighting scheme is carefully designed to adopt adaptive shrinkage estimation principles similar to those underlying the adaptive LASSO (Zou, 2006) (see also Section S2 in the supplement for a detailed discussion of the insights and connections). Specifically, we assign a uniform weight of $n/\log n$ to all coordinates in $\hat{S}$ and a weight of 1 to the remaining coordinates, i.e.,

$$\hat{w}_i = \frac{n}{\log n} \cdot \mathbb{I}(i \in \hat{S}) + \mathbb{I}(i \in \hat{S}^c). \quad (2.7)$$

We emphasize that, given the inherent uncertainty of auxiliary data sources, we do not attempt to fine-tune individual weights within $\hat{S}$ as done in methods such as Cai et al. (2022). Instead, using identical weights of $n/\log n$ within $\hat{S}$ not only simplifies implementation but also enhances robustness by reducing sensitivity to the substantial noise in the auxiliary data or mismatches in supports between target and source parameters.

2. The robustness theory. The theoretical robustness of the proposed estimator (2.5) with prescribed weights $\{\hat{w}_i\}_{i=1}^n$ hinges on the underlying sparsity structure, a requirement formalized in the next condition.

Condition 1. *Let $b_1 > 0, b_2 > 0$ and $b_3 \in (0, 1)$ be arbitrary constants. Suppose that the sparsity level satisfies $k_n/n \in [n^{-1}(\log n)^5, n^{-b_3}]$. The FDR level in (2.6) is chosen such that $q \in [b_1/\log n, b_2/(\log n)^{b_3}]$.*

Remark 1. *In Condition 1, the mild sparsity condition on the rate of k_n/n has been adopted in the analysis of FDR thresholding estimators (Abramovich et al., 2006). Moreover, this condition requires selecting an appropriate FDR level q for the thresholding estimator in (2.6) to ensure effective sparsity adaptation. We have carefully chosen q , which vanishes sufficiently slowly; this choice aligns with the standard requirement for support recovery in FDR-based methods (cf. Abramovich et al., 2006; Meinshausen and Rice, 2006; Cai and Sun, 2017).*

To illustrate how the size of $\hat{S}$ affects the robustness of SATL and to motivate our subsequent data-driven procedure, we first consider an oracle setting in which only the true sparsity level k_n is known. Suppose an oracle provides a candidate support set $S_{\text{OR}} \subseteq [n]$. Using the prototype weighting scheme described above, we construct the corresponding estimator $\hat{\mu}_{\text{OR}}$ via (2.5). Here, S_{OR} is not required to contain the true support S , nor is it assumed to provide an accurate estimate of S . Instead, S_{OR} may be arbitrary, possibly containing many incorrect locations or missing a substantial number of true signals. The

following theorem shows that as long as the size of the candidate support set is properly controlled, the resulting estimator remains robust. Specifically, even in the worst-case scenario where S_{OR} is completely uninformative, the oracle procedure still achieves a near-minimax risk of order $O(k_n \log n)$ over sparse parameter spaces.

Theorem 1. *[Robustness of the oracle procedure with fixed cardinality of candidate support]* Let $\Theta(k_n) = \{\boldsymbol{\mu} \in \mathbb{R}^n : \|\boldsymbol{\mu}\|_0 \leq k_n\}$ denote the k_n -sparse parameter space. Consider the oracle procedure $\hat{\boldsymbol{\mu}}_{\text{OR}}$ constructed from a pre-specified candidate support set S_{OR} satisfying $|S_{\text{OR}}| = O(k_n \log n)$. If Condition 1 holds, then

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} E_{\boldsymbol{\mu}} (\|\hat{\boldsymbol{\mu}}_{\text{OR}} - \boldsymbol{\mu}\|^2) = O(k_n \log n), \quad n \rightarrow \infty.$$

Remark 2. *The requirement that the cardinality of S_{OR} be of order $k_n \log n$ is notably mild, as it permits the use of a conservatively large candidate set. Although our theorem considers an idealized setting, it offers critical insight for designing the data-driven SATL procedure. In practice, we obtain $\hat{S}$ through a screening procedure based on auxiliary data. The inclusion of a null location in $\hat{S}$ constitutes a false positive (FP). The theorem indicates that the screening strategy must explicitly limit the cardinality of $\hat{S}$, or equivalently, the number of false positives in $\hat{S}$, to avoid substantial performance deterioration.*

3 The Data-Driven SATL Algorithm

This section first develops a data-driven SATL algorithm, then discusses conditions under which the algorithm is robust against negative transfer, and finally discusses different conditions under which SATL obtains efficiency gain.

3.1 Theoretical Framework and Overview

We present a high-level overview of the theoretical foundation of the SATL approach. This overview delineates the objectives of the main theorems and elucidates the core princi-

ples that guide our methodological development.

Our theoretical analysis proceeds in two stages, corresponding to the dual objectives of robustness and efficiency. First, in Section 3.2, we develop a data-driven screening procedure that constructs a candidate support set from the external data. Our primary goal is robustness: SATL should remain reliable and avoid negative transfer even when the external datasets are of poor quality. In particular, Remark 2 highlights that controlling the number of false positives in $\hat{S}$ is essential for maintaining stable performance. This observation motivates the use of the BH procedure in the screening step to achieve FDR control. Under mild structural conditions, Theorem 1' establishes that the estimation error attains a near-minimax risk bound even in worst-case transfer scenarios, thereby making the proposed procedure “safe”.

Having established robustness, we investigate the impact of the quality of external data on estimation efficiency in Section 3.3. When the estimated support asymptotically encompasses the true signal set, Theorem 2 shows that SATL achieves a faster convergence rate than the classical minimax benchmark. Even when the external support captures only part of the true signal set, Theorem 3 demonstrates that meaningful efficiency gains can still be obtained.

Collectively, the theory provides a coherent framework in which robustness is guaranteed under weak structural assumptions, and efficiency gains are attainable when the external data are informative.

3.2 Data-Driven Algorithm and Robustness Theory

We prioritize making the procedure robust to external datasets of arbitrary quality, while allowing improved performance when the external datasets are informative. Motivated by this objective and the robustness property of the oracle SATL established in Theorem 1, we first construct candidate support sets from each external dataset through an FDR-controlled screening step.

For each external dataset, we compute coordinatewise p -values using standard approaches

and apply the BH procedure at level $q' \in (0, 1/2)$. The resulting screened support set is

$$\hat{S}^{(m)} = \{i \in [n] : |y_i^{(m)}| \geq t_{BH}^{(m)}\}, \quad (3.8)$$

where $t_{BH}^{(m)} = \tilde{\Phi}^{-1}\left(\frac{\ell^{(m)}q'}{2n}\right)$ and $\ell^{(m)}$ is calculated via (2.3). The role of FDR control here differs from that in the adaptive FDR-thresholding procedure (2.6). Here, the BH procedure is primarily introduced to control the number of false positives, and hence the size of the candidate support set, which is essential for establishing the robustness property in Theorem 1. By preventing excessive false discoveries in the screened supports, the procedure stabilizes the transfer process and reduces the risk of assigning large weights to null coordinates. At the same time, when the external datasets are informative, the screened supports still retain useful structural information and can therefore facilitate effective knowledge transfer.

The final estimator $\hat{S}$ is obtained through the following aggregation scheme: $\hat{S} = \{i \in [n] : c_i \geq p\}$, where c_i denotes the following empirical frequency: $c_i = \frac{1}{M}|\{m \in [M] : i \in \hat{S}^{(m)}\}|$. The parameter $p \in [0, 1]$ controls the degree of agreement required for a coordinate to enter the aggregated support. Larger values of p lead to a more conservative transfer rule, which may provide additional protection against misleading auxiliary information but can also weaken the transfer effect by excluding potentially informative coordinates. In our theoretical analysis and practical implementation, we adopt the default choice $p = 1/M$, corresponding to the union of the screened supports $\hat{S}^{(m)}$. Under the mild conditions introduced below, this choice enables broad transfer of sparsity information while preserving robustness against negative transfer. The complete SATL procedure is summarized in Algorithm 1.

Remark 3. *The parameter q' stabilizes support screening through FDR control. Any fixed choice of $q' \in (0, 1/2)$ provides effective control of the size of the screened supports $\hat{S}^{(m)}$ (Abramovich et al., 2006), and our theoretical results below show that the value of q' has a limited influence on the final risk. In practice, we may take any fixed $q' \in (0, 1/2)$ to implement SATL.*

Our theoretical analysis shows that the choice $p = 1/M$ effectively exploits structural information from informative external datasets while still avoiding negative transfer under mild conditions. A sensitivity analysis for these hyperparameters is provided in Section S7.

Algorithm 1 SATL

Input: $\{y_i : i \in [n]\}$, $\{y_i^{(m)} : i \in [n], m \in [M]\}$, $q \in (0, 1/2)$, $p \in (0, 1)$, $q' \in (0, 1)$, $\varepsilon \in (0, 1)$.

Output: $\hat{\boldsymbol{\mu}} = \{\hat{\mu}_i : i \in [n]\}$

- 1: Run BH procedure with FDR level q' to each external dataset and obtain the rejection sets as the estimated supports $\hat{S}^{(m)}$ as in (3.8).
 - 2: Obtain estimated support $\hat{S}$ by $\hat{S} = \{i \in [n] : |\{m \in [M] : i \in \hat{S}^{(m)}\}|/M \geq p\}$.
 - 3: Obtain weights by $\hat{w}_i = \frac{n}{\log n} \mathbb{I}\{i \in \hat{S}\} + \mathbb{I}\{i \in \hat{S}^c\}$, $i \in [n]$.
 - 4: Use target data to obtain ℓ by applying the BH procedure (2.3) and set thresholds $\{t(\hat{w}_i, \ell, q) : i \in [n]\}$ according to (2.6).
 - 5: Obtain the final estimate $\hat{\boldsymbol{\mu}} = (\hat{\mu}_i : i \in [n])$ using (2.5).
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We next establish the robustness theory for Algorithm 1. We first state a condition on the sparsity structure of external datasets.

Condition 2. For each $m \in [M]$, $k_n^{(m)} := |S_n^{(m)}| = O(k_n \log n)$ as $n \rightarrow \infty$.

Remark 4. The SATL framework entails a data pre-processing stage where baseline correction is applied to remove systematic offsets, thereby isolating task-specific sparse effects and yielding sparse parameter vectors for both the target and auxiliary datasets. Condition 2 requires that the sparsity level of each external dataset be at most slightly higher than that of the target. Practically, the non-null proportion in each dataset can be estimated using well-established methods such as those described in Storey (2002) and Jin and Cai (2007); external datasets whose estimated sparsity deviates markedly from the target can then be filtered out.

The following theorem parallels the oracle result in Theorem 1 and establishes the robustness of the data-driven SATL algorithm.

Theorem 1' [Robustness of the data-driven SATL algorithm.] Under Conditions 1 and 2, for any $p \in (0, 1)$, the estimate $\hat{\boldsymbol{\mu}}$ from Algorithm 1 satisfies: $\sup_{\boldsymbol{\mu} \in \Theta(k_n)} E_{\boldsymbol{\mu}} (\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2) = O(k_n \log n)$, $n \rightarrow \infty$.

Remark 5. *In contrast to existing methods (Li et al., 2021, 2024; He et al., 2024; Lin et al., 2024), which impose explicit bounds on $\|\boldsymbol{\mu} - \boldsymbol{\mu}^{(m)}\|$, the assumptions required by SATL are considerably weaker, allowing for substantial disparities between external and target parameter values. This flexibility enables SATL to accommodate a wide range of source data while maintaining robust performance guarantees.*

3.3 Efficacy of SATL

Having established the robustness of SATL, which hinges on controlling the size of $\hat{S}$ to order $O(k_n \log n)$ (Theorem 1'), we now show that SATL achieves a convergence rate faster than the conventional minimax benchmark by incorporating external data through adaptive weighting. We formally establish conditions under which such efficiency gains are attained. We begin by imposing a mild condition on the sparsity levels of the external datasets, requiring that each external parameter is at most of the same order of sparsity as the target parameter.

Condition 2' For each $m \in [M]$, $k_n^{(m)} \leq C_M k_n$, for some constant $C_M > 0$.

Remark 6. *Although stronger than Condition 2, this condition is still mild and widely applicable. By bounding the number of false positives (null locations in $\hat{S}$), it limits the risk contribution of null coordinates, allowing the estimator to achieve a sharper convergence rate than the conventional minimax benchmark.*

We now introduce a beta-min condition commonly employed in high-dimensional statistics (see, e.g., Bühlmann and Van De Geer, 2011). Define $\mu_{\gamma_m}^{(m)} = C\sqrt{2\log n} + \left\{2\log \frac{2n}{q'(1-\gamma_m)k_n^{(m)}}\right\}^{1/2}$, where $\gamma_m \in [0, 1)$ and $C > \sqrt{2 - b_3}$, with b_3 taken from Condition 1. For each $m \in [M]$, let $S_{\gamma_m}^{(m)} = \{i \in S^{(m)} : |\mu_i^{(m)}| \geq \mu_{\gamma_m}^{(m)}\}$. The parameter γ_m governs the stringency of the beta-min condition on $S_{\gamma_m}^{(m)}$, effectively controlling how many signals in the m -th dataset are difficult to screen. Finally, define $A^M = \cup_{m=1}^M S_{\gamma_m}^{(m)}$ as the union of signals that are reliably detectable across all external datasets.

Condition 3. *The source data satisfy $|S \cap (A^M)^c| \leq C'k_n/(1 + \log(2n)/\log \log n)$, for any constant $C' > 0$.*

Condition 3 requires that most of S be covered by the union of detectable signals from the external datasets, while the uncovered portion is of smaller order than k_n . This ensures that external structural information can be effectively incorporated through the weight construction in (2.7).

Notably, Condition 3 imposes no direct restriction on the individual values of γ_m . Instead, it allows flexible configurations of γ_m as long as sufficient support information is available from the external datasets. For example, in RNA-seq studies, genes active under one biological condition often remain active in related conditions, leading to substantial overlap in signal supports despite differences in expression magnitudes. In this case, most γ_m may be small, so that signals from many external datasets are detectable and together provide good coverage of S . Alternatively, when auxiliary data are collected under multiple treatments of the same sample, good coverage of S can still be achieved as long as datasets from biologically similar treatments provide detectable signals with similar support structures, while the remaining datasets may have arbitrarily large γ_m .

Compared with existing transfer-learning frameworks, Condition 3 accommodates a broader range of heterogeneous settings. In both scenarios above, the conventional value-similarity assumption is violated, making existing transfer-learning methods (Li et al., 2021, 2024; Lin et al., 2024; He et al., 2024) unsuitable. In contrast, SATL relies only on shared structural information and therefore remains applicable in these heterogeneous settings.

Theorem 2 below shows that leveraging informative external data leads to improved convergence rate in estimator risk.

Theorem 2. *Under Conditions 1, Condition 2' and 3, the estimator $\hat{\mu}$ from Algorithm 1*

with $p = 1/M$ satisfies:

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} E_{\boldsymbol{\mu}} (\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2) \leq \max\{(C_1 + C_2 q_n + 3MC_4 C_M)k_n + C_5 C' k_n \log \log n, \\ q_n + 3MC_6 C_M k_n + (C_3 + 2C')k_n \log \log n\},$$

for large enough n , where $C_1 = 1 + c^2$, $C_2 \in (0, 2]$, $C_3 \in (0, 9]$, $C_4 = 4\phi(c)c^{-3}(1 + 1.5c^{-2})$, $C_5 \in (0, 2]$, $C_6 = e^{-c^2/2}$ are positive constants, in which $c = \tilde{\Phi}^{-1}((1 - \varepsilon)/2)$.

Remark 7. The maximum in the risk bound arises from two different regimes of signal strength. When the signals are sufficiently strong, the threshold level ℓ selected by the BH procedure provides an accurate estimate of the sparsity level k_n , leading to the sharper bound given by the first term in the maximum. In contrast, when the signals are weaker, the estimate ℓ may deviate substantially from k_n , resulting in the relatively looser bound represented by the second term. Taking the maximum therefore guarantees uniform risk control over both regimes. Nevertheless, under conditions of Theorem 2, the overall risk bound is of order $O(k_n \log \log n)$, which can be substantially smaller than the classical minimax rate $O(k_n \log(n/k_n))$. This improvement demonstrates the benefit of transferring structural information from the supports of the external datasets.

Remark 8. In all theoretical analyses of this paper, the number of external datasets M is treated as fixed. However, the upper bounds in our theoretical results suggest that M may also increase with n , provided that its growth rate is slower than $\log \log n$. This observation indicates the potential of SATL to accommodate an increasing number of external datasets. A more systematic study of the effect of M is left for future work.

Next, we refine our theory by employing a condition that is strictly weaker than Condition 3.

Condition 4. For some $\eta \in (0, 1)$, $|S \cap (A^M)^c| \leq \eta k_n$.

Theorem 3 below shows that SATL can still achieve efficiency gains under this weaker condition.

Theorem 3. *Under Conditions 1, Condition 2' and 4, the estimator $\hat{\boldsymbol{\mu}}$ from Algorithm 1 with $p = 1/M$ satisfies:*

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} E_{\boldsymbol{\mu}} (\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2) \leq \max\{(C_1 + C_2 q_n + 3MC_4 C_M)k_n + C_5 \eta k_n \log(n/k_n), \\ q_n + 3MC_6 C_M k_n + C_3 k_n \log \log n + 8\eta k_n \log n\},$$

where the constants are the same as those in Theorem 2.

Remark 9. Condition 4 is more practical and spans a continuum of regimes that capture how the informativeness of auxiliary data influences the efficacy of SATL. Specifically, when η is sufficiently small (e.g., $\eta = o(1)$), the estimation risk attains the sharp rate established in Theorem 2. In contrast, as $\eta \rightarrow 1$, the dominant term reduces to $k_n \log n$ —coinciding with the robust-minimax rate given in Theorem 1.

4 Empirical Experiments

This section investigates the empirical performance of SATL and compares it with state-of-the-art alternatives. The following methods are implemented: (a) ABDJ (baseline): the estimator from Abramovich et al. (2006) that only uses the target dataset; (b) SATL: the proposed estimator in Algorithm 1 with $q = q' = 0.1$ and $\varepsilon = 0.01$, where the choice of q is motivated by Condition 1 (in the order of $1/\log n$), while the choice of q' follows the discussion in Remark 3; (c) TransLasso (Li et al., 2021): a two-step transfer learning algorithm with external aggregation, a collection of estimators are constructed and aggregated together; (d) PTL (Lin et al., 2024): a transfer learning method based on the approximate-linear assumption between the source and target parameters; (e) F-AdaTrans and S-AdaTrans (He et al., 2024): adaptive transfer learning methods which can detect and aggregate the feature-

wise (F-AdaTrans) or sample-wise (S-AdaTrans) transferable structures by employing a fused penalty. To facilitate a meaningful and informative comparison across various methods, when implementing transfer learning methods, we set design matrix to be the identity matrix.

We investigate how the degree of informativeness of the external datasets affects transfer learning performance. More supportive numerical results are presented in Section S3 of Supplementary Materials.

The target and external datasets are generated i.i.d. from the model (2.2) with variances to be one. We fix $n = 1000$, target sparsity level $k_n = 40$, and number of external datasets $M = 10$. The target parameter is specified as follows: $(\mu_1, \dots, \mu_{k_n})$ take values evenly spaced between 0.1 and 10, while the remaining μ_i for $i > k_n$ are set to zero. For the external datasets, we first specify the support sizes as $(k^{(1)}, \dots, k^{(M)}) = (30, 36, 42, 47, 53, 58, 64, 69, 75, 80)$. We then determine the size of overlap between each $\boldsymbol{\mu}^{(m)}$ and $\boldsymbol{\mu}$ by enforcing $|S \cap S^{(m)}| = \min\{\lceil r_m k_n \rceil, k^{(m)}\}$, where $r_m \in (0, 1)$ is a *similarity ratio* quantifying the structural similarity between the m -th external dataset and the target dataset. Given this constraint, we randomly select $|S \cap S^{(m)}|$ coordinates from S and $k^{(m)} - |S \cap S^{(m)}|$ coordinates from S^c to form the support $S^{(m)}$. Given the generated supports of the external parameters, the true parameters of the m -th source data are assigned as follows: (a) For signals in $S \cap S^{(m)}$: we set $\mu_i^{(m)} = \mu_i + N(d, 0.1^2)$; (b) For signals in $S^{(m)} \setminus S$: the values are evenly spaced between 0.1 and 20, with $|S^{(m)} \setminus S| - 1$ equal steps. Once generated, the parameters $\{\boldsymbol{\mu}^{(m)}\}_{m=1}^M$ remain fixed across all replications.

We compare the performance of methods under varying levels of d and similarity ratio r_m . Specifically, we set $r_m = r$ for all m and consider $d \in \{0.5, 7, 13.5, 20\}$, $r \in \{0.05, 0.1, 0.15, 0.25, 0.5, 0.8\}$. The performance is evaluated using the average estimation error $\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2/n$. Each simulation setting is replicated over 200 independent trials.

The performance of all methods under these settings is summarized in Figure 1, where the x -axis represents the dissimilarity ratio defined as $(1 - r)$. SATL demonstrates substan-

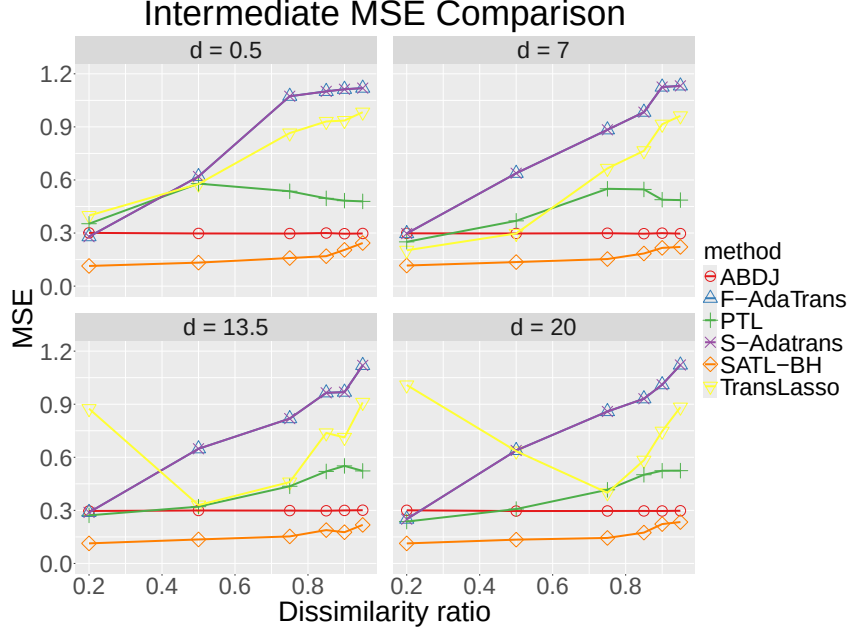


Figure 1: Estimation performance of different methods.

tial efficiency gains over both existing transfer learning approaches and ABDJ. In particular, when similarity decreases and the difference d increases, most transfer learning methods perform worse than our proposed method. This is mainly because these methods either discard external datasets with low similarity or rely on linear combinations of external parameters, which cannot approximate the target parameter well when the external datasets are dissimilar. Specifically, when $d = 0.5$, the common condition ($\|\boldsymbol{\mu} - \boldsymbol{\mu}^{(m)}\|$ is small) in existing transfer learning methods is satisfied (Li et al., 2021; He et al., 2024). Similar external datasets can be integrated by these methods, yielding low MSE for large r . When $d \geq 7$, even external datasets with large r become uninformative for transfer learning methods, so the performance depends on both d and r and does not exhibit a clear trend.

By contrast, SATL exploits the sparsity structure regardless of the value of d . Moreover, SATL incorporates support information from external datasets even when each dataset individually shares limited similarity with the target parameter. This observation is consistent with Theorem 2. When r is small, the coverage of S decreases, and the estimation risk approaches that of the baseline ABDJ method. This aligns with Theorem 3.

5 Real data analysis

In this section, we compare performance of methods using an RNA-seq dataset. The dataset comprises fifteen groups corresponding to different experimental conditions, and is publicly available at the website presented in Section S8 of Supplementary Materials.

The summary of expression data is shown in Table S.1 of Supplementary Materials. In each group, n genes are measured, each with three replicate samples. The experimental conditions differ in medication, organ, and observation time, exhibiting different expression levels. Conditions with similar organs share structural similarities, as illustrated in heatmaps in Supplementary Materials. Our goal is to estimate expression levels μ of n genes in the group *Acute Colon Control*. The other groups serve as external data, which involve different organs but may contain informative patterns, since these organs are anatomically adjacent and functionally related.

We apply different methods to two replicates from *Acute Colon Control*, and evaluate performance by predicting the expression levels of the remaining replicate. The transfer learning methods considered in Section 4 are excluded as they require substantial memory and computational resources to handle large matrices and are not feasible for this dataset. The target summary statistics are computed from the two replicates with adjusted variance set to one. The external observations $\mathbf{y}^{(m)}$ correspond to the mean expression levels of all groups except *Acute Colon Control*, with their variances also adjusted to one. To assess the contribution of external datasets, we consider four settings: (a) All external groups are included; (b) Only *Acute Colon Imeglimin* and *Acute Colon Metformin* are included; (c) Only groups without medication are included; and (d) Only *Chronic Colon Control* is included. Intuitively, the amount of useful information from external datasets decreases from Setting (a) to Setting (d), as fewer datasets share similar organs or treatments. The hyper-parameters in SATL are chosen in the same manner as in Section 4. The results are summarized in Table 1.

Table 1: Summary of results

Method	Error(Setting (a))	Error(Setting (b))	Error(Setting (c))	Error(Setting (d))
ABDJ	1.140	1.140	1.140	1.140
SATL	0.650	0.683	0.671	0.712

These results demonstrate that the sparsity structure in source data can be effectively transferred by SATL, leading to improved performance compared with conventional sparse estimation approaches. Moreover, datasets that are more similar to the target yield better results, due to more accurate identification of the target support.

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Electronic companion (Appendices) to

“Transfer Learning for Sparse Estimation: A Structure-Adaptive Approach”

This Supplement is organized as follows. Section S1 provides the technical proofs for all theories. Section S2 presents intuitions of construction of weights in (2.7). Section S3 provides more numerical results of comparison of performance from different methods. Finally, Section S8 presents more information for RNA-seq data in Section 5.

S1 Proofs

In this section we provide all the details of proofs of theoretical results, including extra auxiliary results about BH procedure and SATL with given weights. Then we prove results about SATL in the following order: Theorem 2, Theorem 3 and Theorem 1, for the sake of narrative convenience.

S1.1 Theoretical results of BH procedure

Our proof is closely related to previous results in Abramovich et al. (2006), and we first review some notations and results on the estimation based on the FDR-thresholding procedure in this section. Recall that ℓ is the cut-off in the BH procedure, and the right tail Gaussian quantiles $t_i = \tilde{\Phi}^{-1}\left(\frac{q_n i}{2n}\right)$ for each $i \in [n]$. Note that we rewrite the FDR level q in SATL as q_n to show its dependence on n . Define the exceedance number $N(t_l) = \#\{i : |y_{(i)}| \geq t_l\}$, where $|y|_{(1)} \geq |y|_{(2)} \geq \cdots \geq |y|_{(n)}$. The mean exceedance function is defined as

$$M(l, \boldsymbol{\mu}) = \mathbb{E}_{\boldsymbol{\mu}} N(t_l) = \sum_{i=1}^n \mathbb{P}_{\boldsymbol{\mu}}(|y_i| \geq t_l).$$

Based on the mean discovery number defined as $k(\boldsymbol{\mu}) := \inf\{l \in \mathbb{R} : M(l, \boldsymbol{\mu}) = l\}$, by

Eq. (5.3) of [Abramovich et al. \(2006\)](#), we have

$$k(\boldsymbol{\mu}) \leq k_n / (1 - q_n(1 + o(1))),$$

for all $\boldsymbol{\mu} \in \Theta(k_n)$.

Next we introduce the theoretical results of bounds of ℓ . There are two important bounding points for the location of ℓ :

$$k_-(\boldsymbol{\mu}) = \begin{cases} k(\boldsymbol{\mu}) - \alpha_n k_n, & k(\boldsymbol{\mu}) \geq 2\alpha_n k_n, \\ 0, & k(\boldsymbol{\mu}) < 2\alpha_n k_n, \end{cases}$$

and

$$k_+(\boldsymbol{\mu}) = k(\boldsymbol{\mu}) \vee \alpha_n k_n + \alpha_n k_n,$$

where $\alpha_n = 1/(b_4 \tau_n)$, with $\tau_n = (2 \log(n/k_n))^{1/2}$ and $b_4 = 1/4$. We say that an event $A_n(\boldsymbol{\mu})$ is $\Theta(k_n)$ -likely if there exist constants c_0, c_1 not depending on n and $\Theta(k_n)$ such that

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} \mathbb{P}_{\boldsymbol{\mu}}\{A_n^c(\boldsymbol{\mu})\} \leq c_0 \exp\{-c_1 \log^2 n\}.$$

With these Preliminaries, [Abramovich et al. \(2006\)](#) showed that

Proposition S.1 (Proposition 5.1 of [Abramovich et al. \(2006\)](#)). *For any q_n such that $q_n \geq b_1/\log n$ and $q_n \rightarrow q \in [0, 1)$, if $k_n/n \in [n^{-1}(\log n)^5, b_2 n^{-b_3}]$, for constants $b_2 > 0$ and $b_3 \in (0, 1)$, then it is $\Theta(k_n)$ -likely that*

$$k_-(\boldsymbol{\mu}) \leq \ell \leq k_+(\boldsymbol{\mu}).$$

Proposition [S.1](#) provides bounds of ℓ and is used to simplify the analysis of weighted FDR thresholding.

S1.2 Theoretical results of SATL with given weights

To show the proofs of main theorems clearly, some lemmas considering SATL with given weights are needed. We first consider a special setting in which all weights are assigned to the correct coordinates. Let $c_n = n/\log n$.

Condition S.1. *Given the weight vector $\{\hat{w}_i\}_{i \in [n]}$, it satisfies $\{i \in [n] : \hat{w}_i = c_n\} = S$.*

Lemma S.1. *Assume Condition 1 holds. Given weights satisfying Condition S.1, the estimator $\hat{\boldsymbol{\mu}}$ in (2.5) satisfies*

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} \mathbb{E}_{\boldsymbol{\mu}} (\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 | \{\hat{w}_i\}_{i \in [n]}) \leq \max\{(C_1 + C_2 q_n)k_n, q_n + C_3 k_n \log \log n\},$$

for large enough n , where $C_1 = 1 + \tilde{\Phi}^{-1}((1 - \varepsilon)/2)$, $C_2 \in (0, 2]$ and $C_3 \in (0, 9]$ are constants.

Lemma S.1 serves as a basic result for proofs of all other theoretical results. The proof is shown in Section S1.3.

Remark S.1. *When proving Lemma S.1, we further require the existence of $\lim_{n \rightarrow \infty} c_n q_n / n$ as a technical regularity condition for convenience. This assumption serves two purposes: first, it corresponds to realistic settings observed in practical applications; second, it eliminates degenerate cases that would otherwise necessitate additional technical machinery in the proof. Similar arguments hold when proving the other theoretical results for convenience.*

We then consider the case when most weights are allocated to the correct coordinates. Note that including a zero (respectively, nonzero) coordinate in $\hat{S}$ corresponds to a false positive (FP) (respectively, true positive (TP)), whereas including a nonzero (respectively, zero) coordinate in $\hat{S}^c$ corresponds to a false negative (FN) (respectively, true negative (TN)). Define $S_{\text{FP}} = \{i \in S^c : \hat{w}_i = c_n\}$, $S_{\text{TP}} = \{i \in S : \hat{w}_i = c_n\}$, $S_{\text{TN}} = \{i \in S^c : \hat{w}_i = 1\}$, and $S_{\text{FN}} = \{i \in S : \hat{w}_i = 1\}$.

Lemma S.2. *Assume Condition 1 holds. Given weight vector $\{\hat{w}_i\}_{i=1}^n$, the estimator $\hat{\boldsymbol{\mu}}$ in (2.5) satisfies*

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} \mathbb{E}_{\boldsymbol{\mu}} \left(\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mid \{\hat{w}_i\}_{i \in [n]} \right) \leq \max\{C_2 k_n q_n + C_4 |S_{\text{FP}}| + C_1 k_n + C_5 |S_{\text{FN}}| \log(n/k_n), \\ q_n + C_6 |S_{\text{FP}}| + C_3 k_n \log \log n + 2 |S_{\text{FN}}| \log(2n/q_n)\},$$

for large enough n , where C_1 , C_2 and C_3 are the same constants as in Lemma S.1, and $C_4 = 4\phi(c)c^{-3}(1+1.5c^{-2})$, $C_5 \leq 2$, $C_6 = e^{-c^2/2}$ are positive constants, in which $c = \tilde{\Phi}^{-1}((1-\varepsilon)/2)$.

Lemma S.2 provides a fundamental result illustrating how the allocation of the weight vector affects estimation performance.

Next, we consider the case where some weights are not correctly matched to the target. We first present a basic result characterizing the effect of the weight vector on the estimation risk.

Lemma S.3. *Assume Condition 1 holds. Given weight vector $\{\hat{w}_i\}_{i=1}^n$, the estimator $\hat{\boldsymbol{\mu}}$ in (2.5) satisfies*

$$\sup_{\boldsymbol{\mu} \in \Theta(k_n)} \mathbb{E}_{\boldsymbol{\mu}} \left(\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mid \{\hat{w}_i\}_{i \in [n]} \right) \leq \max\{C_2 k_n q_n + C_4 |S_{\text{FP}}| + C_1 |S_{\text{TP}}| + C_5 |S_{\text{FN}}| \log(n/k_n), \\ q_n + C_6 |S_{\text{FP}}| + C_3 |S_{\text{TP}}| \log \log n + 8 |S_{\text{FN}}| \log n\},$$

for large enough n , where constants are the same as in Lemma S.2.

S1.3 Proof of Lemma S.1

For notational simplicity, we suppress the condition that the weights $\{\hat{w}_i\}_{i=1}^n$ are regarded as fixed when taking expectations. We will use following notations of some constants. Recall that $\alpha_n = 1/(b_4 \tau_n)$ with $b_4 = 1/4$, and $\tau_n = (2 \log(n/k_n))^{1/2}$. Let $a_n = O(b_n)$ denote $|a_n/b_n| \leq c < \infty$ for some constant c when n is large enough. Note that the proposed threshold is $t(w_i, \ell, q_n)$, where we do not use $\hat{w}$ to emphasize that weights are given. Recall

that $\hat{\mu}(y, t) = y + \delta(y, t)$ and let $\hat{\mu}_i(t) = \hat{\mu}(y_i, t)$. Denote $r_S(\lambda, \mu) = \mathbb{E}_\mu[\hat{\mu}(x, \lambda) - \mu]^2$ as the risk for soft-thresholding estimator with $x \sim \mathcal{N}(\mu, 1)$ and threshold λ . The proof sketch of Lemma S.1 is as follows:

Step 1: We show that ℓ is close to the mean discovery number $k(\mu)$ with high probability. The proof highly depends on the properties of the mean detection function with threshold $t(w_i, \ell, q_n)$.

Step 2: We consider two cases of ℓ . First, when $2\alpha_n k_n \leq \ell \leq (\frac{1}{1-q_n/2} + \alpha_n)k_n$, the proof is straightforward since ℓ is close to k_n . Second, when $1 \leq \ell \leq 2\alpha_n k_n$, we can show that most non-zero signals are below threshold $t(w_i, 2\alpha_n k_n, q_n)$. Then the estimator can achieve the desired property since most μ_i are small.

Proof. Let $A_n = \{k_-(\mu) \leq \ell \leq k_+(\mu)\}$ represent the event that the cut-off ℓ lies in the interval $[k_-(\mu), k_+(\mu)]$. By Proposition S.1, event A_n is $\Theta(k_n)$ -likely. Using Lemma 8.4 and Lemma 12.3 of Abramovich et al. (2006), we have

$$\begin{aligned}
& \mathbb{E}_\mu \|\hat{\mu} - \mu\|^2 \mathbb{I}\{A_n^c\} \\
& \leq 4nc_0 [t(\min_i w_i, 1, q_n)]^2 \exp(-c_1 \log^2 n) \\
& \leq c_0 n \left(2 \log \frac{2n}{\min_i w_i q_n} - \log \log \frac{2n}{\min_i w_i q_n} - c_2 \right) \exp(-c_1 \log^2 n) \\
& = o(1),
\end{aligned} \tag{S1.1}$$

where we use the fact that $\min_i w_i = 1$. Thus, in the following proof, it suffices to analyze the term $\mathbb{E}_\mu \|\hat{\mu} - \mu\|^2 \mathbb{I}\{A_n\}$, which is discussed partly into two cases according to the relation of $k(\mu)$ with $2\alpha_n k_n$.

Case 1: When $k(\mu) \geq 2\alpha_n k_n$.

Since $k_+(\mu) = k(\mu) \vee \alpha_n k_n + \alpha_n k_n$, when n is large, in the event A_n we have

$$\alpha_n k_n \leq \ell \leq \frac{1}{1 - q_n/2} k_n + \alpha_n k_n := \beta_n k_n.$$

By the fact that $t(w, \ell, q)$ is monotone in ℓ , we have that $t(w_i, \ell, q_n)$ is between $t(w_i, \beta_n k_n, q_n)$ and $t(w_i, \alpha_n k_n, q_n)$ in the event A_n . Note that $[\hat{\mu}(x, \lambda) - \mu]^2$ is first decreasing and then increasing in λ , which implies that $[\hat{\mu}(x, \lambda) - \mu]^2$ achieves the maximum values at the boundary points of λ . Thus, in the event A_n we have

$$\begin{aligned} & (\hat{\mu}_i[t(w_i, \ell, q_n)] - \mu_i)^2 \\ & \leq \max \{ (\hat{\mu}_i[t(w_i, \beta_n k_n, q_n)] - \mu_i)^2, (\hat{\mu}_i[t(w_i, \alpha_n k_n, q_n)] - \mu_i)^2 \} \\ & \leq (\hat{\mu}_i[t(w_i, \beta_n k_n, q_n)] - \mu_i)^2 + (\hat{\mu}_i[t(w_i, \alpha_n k_n, q_n)] - \mu_i)^2. \end{aligned}$$

Therefore, we have

$$\begin{aligned} & \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} \\ &= \sum_{i=1}^n \mathbb{E}_{\boldsymbol{\mu}} (\hat{\mu}_i[t(w_i, \ell, q_n)] - \mu_i)^2 \mathbb{I}\{A_n\} \\ &\leq \sum_{i=1}^n \mathbb{E}_{\boldsymbol{\mu}} [(\hat{\mu}_i[t(w_i, \beta_n k_n, q_n)] - \mu_i)^2 + (\hat{\mu}_i[t(w_i, \alpha_n k_n, q_n)] - \mu_i)^2] \\ &= \sum_{i=1}^n r_S(t(w_i, \beta_n k_n, q_n), \mu_i) + \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i). \end{aligned} \tag{S1.2}$$

We will utilize the same technique to upper bound the last two terms in (S1.2), so only the term $\sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i)$ is discussed below. For this term, we split the term into two parts:

$$\begin{aligned} & \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ &= \sum_{i: \mu_i=0} r_S(t(w_i, \alpha_n k_n, q_n), 0) + \sum_{i: \mu_i \neq 0} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i). \end{aligned}$$

(i) For the individuals with $\mu_i = 0$, according to Eq. (8.5) of [Johnstone \(2011\)](#), we have $r_S(\lambda, 0) \leq 4\lambda^{-3}(1 + 1.5\lambda^{-2})\phi(\lambda)$ for all $\lambda > 0$, and thus $r_S(\lambda, 0) \sim 4\lambda^{-3}\phi(\lambda)$ when $\lambda \rightarrow \infty$.

By the weights construction, we have $w_i = 1$ for all i such that $\mu_i = 0$, which yields

$$\begin{aligned} & r_S(t(w_i, \alpha_n k_n, q_n), 0) \\ & \leq 4 (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)). \end{aligned}$$

Note that $\frac{\alpha_n k_n q_n}{n} \rightarrow 0$, as $n \rightarrow \infty$. Hence,

$$t(1, \alpha_n k_n, q_n) = \tilde{\Phi}^{-1} \left(\frac{1}{2} \left(\frac{\alpha_n k_n}{n} q_n \wedge (1 - \varepsilon) \right) \right) \rightarrow \infty.$$

Since $\phi(\lambda) \sim \lambda \tilde{\Phi}(\lambda)$ as $\lambda \rightarrow \infty$, we have

$$\begin{aligned} & \sum_{i: \mu_i=0} r_S(t(w_i, \alpha_n k_n, q_n), 0) \\ & \leq 4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)) \\ & \simeq 4n (t(1, \alpha_n k_n, q_n))^{-2} (1 + o(1)) \frac{1}{2} \left(\frac{\alpha_n k_n q_n}{n} \wedge (1 - \varepsilon) \right) \\ & \leq 2(1 + o(1)) (t(1, \alpha_n k_n, q_n))^{-2} \alpha_n k_n q_n \\ & = 2(1 + o(1)) \frac{\alpha_n k_n q_n}{2 \log \left(2 \left(\frac{n}{\alpha_n k_n q_n} \vee \frac{1}{1-\varepsilon} \right) \right) - \log \log \left(2 \left(\frac{n}{\alpha_n k_n q_n} \vee \frac{1}{1-\varepsilon} \right) \right) - c_2} \\ & = (1 + o(1)) \frac{\alpha_n k_n q_n}{\log \left(\frac{2n}{\alpha_n k_n q_n} \right) (1 + o(1))} \\ & \leq C_2 k_n q_n, \end{aligned}$$

for some constant $0 < C_2 \leq 2$ when n is large enough.

(ii) Consider the term $r_S(t(w_i, \alpha_n k_n, q_n), \mu_i)$ for all i such that $\mu_i \neq 0$. According to Lemma 8.3 of [Johnstone \(2011\)](#), $r_S(\lambda, \mu) \leq 1 + \lambda^2$ for all $\lambda > 0$ and $\mu \in \mathbb{R}$. When n is large

enough, we have

$$\begin{aligned}
& \sum_{i:\mu_i \neq 0} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
& \leq \sum_{i:\mu_i \neq 0} [1 + (t(w_i, \alpha_n k_n, q_n))^2] \\
& = k_n [1 + (t(c_n, \alpha_n k_n, q_n))^2] \\
& \leq C_1 k_n,
\end{aligned}$$

for constant $C_1 = 1 + (\tilde{\Phi}^{-1}((1-\varepsilon)/2))^2$, where the last step we use the fact that $c_n \alpha_n k_n q_n / n \rightarrow \infty$ and thus $t(c_n, \alpha_n k_n, q_n) = \tilde{\Phi}^{-1}((1-\varepsilon)/2)$ which is a constant. Note that the analysis for the term involving $\beta_n k_n$ is the same, and we omit it.

Combining the results above, for all $\boldsymbol{\mu} \in \Theta(k_n)$, we have

$$\begin{aligned}
& \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \\
& = \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} + \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\} \\
& \leq o(1) + \sum_{i=1}^n r_S(t(w_i, \beta_n k_n, q_n), \mu_i) + \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
& \leq (C_1 + C_2 q_n) k_n,
\end{aligned}$$

for large enough n .

Case 2: When $k(\boldsymbol{\mu}) \leq 2\alpha_n k_n$.

In this case, we have $1 \leq \ell \leq k(\boldsymbol{\mu}) + \alpha_n k_n \leq 3\alpha_n k_n$ in the event A_n , and hence

$$\begin{aligned}
& (\hat{\mu}_i(t(w_i, \ell, q_n)) - \mu_i)^2 \\
& \leq \max \{ [\hat{\mu}_i(t(w_i, 1, q_n)) - \mu_i]^2, [\hat{\mu}_i(t(w_i, 3\alpha_n k_n, q_n)) - \mu_i]^2 \} \\
& \leq [\hat{\mu}_i(t(w_i, 1, q_n)) - \mu_i]^2 + [\hat{\mu}_i(t(w_i, 3\alpha_n k_n, q_n)) - \mu_i]^2
\end{aligned}$$

Therefore we have

$$\begin{aligned}
& \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} \\
&= \sum_{i=1}^n \mathbb{E}_{\boldsymbol{\mu}} (\hat{\mu}_i[t(w_i, \ell, q_n)] - \mu_i)^2 \mathbb{I}\{A_n\} \\
&\leq \sum_{i=1}^n \mathbb{E}_{\boldsymbol{\mu}} [(\hat{\mu}_i[t(w_i, 1, q_n)] - \mu_i)^2 + (\hat{\mu}_i[t(w_i, 3\alpha_n k_n, q_n)] - \mu_i)^2] \\
&= \sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i) + \sum_{i=1}^n r_S(t(w_i, 3\alpha_n k_n, q_n), \mu_i),
\end{aligned} \tag{S1.3}$$

the expectation of squared error is the same when focusing on the term involving $3\alpha_n k_n$. Therefore, we concentrate on the term $\sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i)$. Devide it into two parts as before:

$$\sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i) = \sum_{i:\mu_i=0} r_S(t(w_i, 1, q_n), 0) + \sum_{i:\mu_i \neq 0} r_S(t(w_i, 1, q_n), \mu_i).$$

(i) For the part $\sum_{i:\mu_i=0} r_S(t(w_i, 1, q_n), 0)$, by Eq. (8.7) of [Johnstone \(2011\)](#), we have

$$\begin{aligned}
\sum_{i:\mu_i=0} r_S(t(w_i, 1, q_n), 0) &\leq \sum_{i:\mu_i=0} \exp\left(-\frac{1}{2}[t(w_i, 1, q_n)]^2\right) \\
&= \sum_{i:\mu_i=0} \exp\left(-\frac{1}{2}[t(1, 1, q_n)]^2\right).
\end{aligned}$$

By Lemma 12.3 of [Abramovich et al. \(2006\)](#), since $q_n/n \rightarrow 0$, we have

$$\sum_{i:\mu_i=0} \exp\left(-\frac{1}{2}[t(1, 1, q_n)]^2\right) = \sum_{i:\mu_i=0} \exp\left\{-\frac{1}{2}\left(2\log\frac{2n}{q_n} - \log\log\frac{2n}{q_n} - c_2\right)\right\}.$$

Therefore, we obtain

$$\begin{aligned}
& \sum_{i:\mu_i=0} r_S(t(w_i, 1, q_n), 0) \\
& \leq \sum_{i:\mu_i=0} \exp \left\{ -\frac{1}{2} \left(2 \log \frac{2n}{q_n} - \log \log \frac{2n}{q_n} - c_2 \right) \right\} \\
& \leq n \exp \left\{ -\log \frac{2n}{q_n} (1 + o(1)) \right\} \\
& \leq q_n,
\end{aligned} \tag{S1.4}$$

for large enough n .

(ii) For the part $\sum_{i:\mu_i \neq 0} r_S(t(w_i, 1, q_n), \mu_i)$, using Lemma 8.3 of [Johnstone \(2011\)](#), we have

$$\sum_{i:\mu_i \neq 0} r_S(t(w_i, 1, q_n), \mu_i) \leq \sum_{i:\mu_i \neq 0} (1 + [t(w_i, 1, q_n)]^2).$$

Since $w_i = c_n$ for all i such that $\mu_i \neq 0$, and $c_n q_n / n \rightarrow 0$, Lemma 12.3 of [Abramovich et al. \(2006\)](#) yields

$$\begin{aligned}
\sum_{i:\mu_i \neq 0} (1 + [t(w_i, 1, q_n)]^2) & \leq k_n (1 + [t(c_n, 1, q_n)]^2) \\
& = k_n \left(1 + 2 \log \frac{2n}{c_n q_n} - \log \log \frac{2n}{c_n q_n} - c_2 \right),
\end{aligned}$$

for some constant $c_2 \in [1.8, 3]$. Therefore, we obtain

$$\begin{aligned}
\sum_{i:\mu_i \neq 0} r_S(t(w_i, 1, q_n), \mu_i) & \leq k_n \left(1 + 2 \log \frac{2n}{c_n q_n} - \log \log \frac{2n}{c_n q_n} - c_2 \right) \\
& = 2k_n \log \left(\frac{2n}{c_n q_n} \right) (1 + o(1)) \\
& \leq C_3 k_n \log \log n,
\end{aligned}$$

for some constant $C_3 \leq 3$, when n is large enough.

Combining the results above, for all $\boldsymbol{\mu} \in \Theta(k_n)$, we have

$$\begin{aligned}
& \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \\
&= \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} + \mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\} \\
&\leq o(1) + \sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i) + \sum_{i=1}^n r_S(t(w_i, 3\alpha_n k_n, q_n), \mu_i) \\
&\leq q_n + C_3 k_n \log \log n,
\end{aligned}$$

for large enough n . Combining the two cases yields the desired result. $\square$

S1.4 Proof of Lemma S.2

The proof of Lemma S.2 follows the same argument as that of Lemma S.1, and thus some details are omitted for brevity.

Proof. The term $\mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\}$ can be handled in the same way as in (S1.1), so we focus on $\mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\}$. The overall proof proceeds similarly to that of Lemma S.1, and some technical details are omitted for brevity.

Case 1: When $k(\boldsymbol{\mu}) \geq 2\alpha_n k_n$.

The analysis follows the same reasoning as in Lemma S.1, except that the summation is now partitioned according to the weights. Formally, we have

$$\begin{aligned}
& \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
&= \sum_{i: \mu_i=0} r_S(t(w_i, \alpha_n k_n, q_n), 0) + \sum_{i: \mu_i \neq 0} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
&= \left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), 0) \\
&\quad + \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i).
\end{aligned}$$

We discuss the null part and non-null part separately.

(i) The risk can be rewritten as follows.

$$\begin{aligned}
& \left(\sum_{j \in S_{\text{TN}}} + \sum_{j \in S_{\text{FP}}} \right) r_S(t(w_j, \alpha_n k_n, q_n), 0) \\
& \leq 4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)) \\
& \quad + 4|S_{\text{FP}}| (t(c_n, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(c_n, \alpha_n k_n, q_n))^{-2}) \phi(t(c_n, \alpha_n k_n, q_n)).
\end{aligned} \tag{S1.5}$$

Since $\alpha_n k_n q_n / n \rightarrow 0$, we have $t(1, \alpha_n k_n, q_n) \rightarrow \infty$, as $n \rightarrow \infty$. Therefore, $\phi(t(1, \alpha_n k_n, q_n)) \sim t(1, \alpha_n k_n, q_n) \tilde{\Phi}(t(1, \alpha_n k_n, q_n))$, which leads to

$$\begin{aligned}
& 4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)) \\
& \simeq 4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \tilde{\Phi}(t(1, \alpha_n k_n, q_n)) \\
& \leq (1 + o(1)) \frac{\alpha_n k_n q_n}{\log\left(\frac{2n}{\alpha_n k_n q_n}\right) (1 + o(1))} \\
& \leq C_2 k_n q_n,
\end{aligned} \tag{S1.6}$$

for the same C_2 as in the proof of Lemma S.1.

For the second term of (S1.5), since $\min\{c_n \alpha_n k_n q_n / n, 1 - \varepsilon\} \leq 1 - \varepsilon$, we have

$$t(c_n, \alpha_n k_n, q_n) \geq \tilde{\Phi}^{-1}((1 - \varepsilon)/2) =: c. \tag{S1.7}$$

Therefore,

$$\begin{aligned}
& 4|S_{\text{FP}}| (t(c_n, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(c_n, \alpha_n k_n, q_n))^{-2}) \phi(t(c_n, \alpha_n k_n, q_n)) \\
& \leq 4|S_{\text{FP}}| c^{-3} (1 + 1.5c^{-2}) \phi(c) = C_4 |S_{\text{FP}}|,
\end{aligned}$$

where $C_4 = 4c^{-3} (1 + 1.5c^{-2}) \phi(c)$. Combining the results above, we have

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), 0) \leq C_2 k_n q_n + C_4 |S_{\text{FP}}|.$$

(ii) For the non-null part, note that $\min_{i \in S_{\text{FN}}} w_i = 1$. The risk can be rewritten as follows.

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq |S_{\text{TP}}| [1 + (t(c_n, \alpha_n k_n, q_n))^2] + |S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2] \\ & \leq C_1 k_n + |S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2], \end{aligned}$$

where C_1 is the same constant as in Lemma S.1. Note that

$$\begin{aligned} (t(1, \alpha_n k_n, q_n))^2 & \lesssim \left(\tilde{\Phi}^{-1} \left(\frac{\alpha_n k_n q_n}{2n} \right) \right)^2 \\ & = 2 \log \left(\frac{2n}{\alpha_n k_n q_n} \right) - \log \log \left(\frac{2n}{\alpha_n k_n q_n} \right) - c \\ & \simeq 2 \log \left(\frac{2n}{\alpha_n k_n q_n} \right). \end{aligned} \tag{S1.8}$$

We have

$$|S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2] \leq C_5 |S_{\text{FN}}| \log(n/k_n), \tag{S1.9}$$

where $C_5 \leq 2$ is a positive constant, when n is large enough.

Finally, we obtain

$$\left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \leq C_1 k_n + C_5 |S_{\text{FN}}| \log(n/k_n),$$

when n is large enough.

Similarly, the same result holds for the term

$$\sum_i r_S(t(w_i, \beta_n k_n, q_n), \mu_i)$$

in this case.

Case 2: When $k(\boldsymbol{\mu}) \leq 2\alpha_n k_n$.

Following (S1.3) from the discussion in Case 2 of the proof of Lemma S.1, we mainly focus on the term $\sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i)$, since the term $\sum_{i=1}^n r_S(t(w_i, 3\alpha_n k_n, q_n), \mu_i)$ can be treated in the same manner as in the case where $k(\boldsymbol{\mu}) \geq 2\alpha_n k_n$. The first term can then be further decomposed as follows.

$$\begin{aligned} & \sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i) \\ &= \sum_{i: \mu_i=0} r_S(t(w_i, 1, q_n), 0) + \sum_{i: \mu_i \neq 0} r_S(t(w_i, 1, q_n), \mu_i) \\ &= \left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \\ & \quad + \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, 1, q_n), \mu_i). \end{aligned}$$

We also discuss the null part and non-null part separately.

(i) Using the same notation as before, we rewrite the risk corresponding to the null part as follows.

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \\ & \leq n \exp \left(-\frac{1}{2} (t(1, 1, q_n))^2 \right) + |S_{\text{FP}}| \exp \left(-\frac{1}{2} (t(c_n, 1, q_n))^2 \right). \end{aligned}$$

By (S1.4), we have $n \exp \left(-\frac{1}{2} (t(1, 1, q_n))^2 \right) \leq q_n$. For the second term, by the same technique in (S1.7), we have

$$t(c_n, 1, q_n) \geq \tilde{\Phi}^{-1}((1 - \varepsilon)/2) =: c.$$

Therefore, we obtain

$$|S_{\text{FP}}| \exp \left\{ -\frac{1}{2} [t(c_n, 1, q_n)]^2 \right\} \leq |S_{\text{FP}}| \exp(-c^2/2) = C_6 |S_{\text{FP}}|,$$

where $C_6 = e^{-c^2/2}$. Combining the results above, we have

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \leq q_n + C_6 |S_{\text{FP}}|,$$

when n is large enough.

(ii) Using the same notation as before, the risk corresponding to the non-null part can be rewritten as

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, 1, q_n), \mu_i) \\ & \leq k_n [1 + (t(c_n, 1, q_n))^2] + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2] \\ & \leq C_3 k_n \log \log n + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2], \end{aligned}$$

where C_3 is the same constant as in Lemma S.1. Note that

$$(t(1, 1, q_n))^2 \lesssim 2 \log \left(\frac{2n}{q_n} \right). \quad (\text{S1.10})$$

Therefore, we obtain

$$\sum_i r_S(t(w_i, 1, q_n), \mu_i) \leq q_n + C_6 |S_{\text{FP}}| + C_3 k_n \log \log n + 2 |S_{\text{FN}}| \log(2n/q_n),$$

when $k(\boldsymbol{\mu}) \leq 2\alpha_n k_n$, for large enough n . The proof is completed by combining the results of two cases.

□

S1.5 Proof of Lemma S.3

Proof. The term $\mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\}$ yields the same result as in (S1.1), so we focus on $\mathbb{E}_{\boldsymbol{\mu}} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\}$. The proof proceeds similarly to that of Lemma S.2, and some details are omitted for brevity. As before, we discuss different cases of $k(\boldsymbol{\mu})$.

Case 1: When $k(\boldsymbol{\mu}) \geq 2\alpha_n k_n$.

The analysis follows the same arguments as in the proof of Lemma S.1, except that the summation must now be partitioned according to the weights. Formally, now we have

$$\begin{aligned}
& \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
&= \sum_{i: \mu_i=0} r_S(t(w_i, \alpha_n k_n, q_n), 0) + \sum_{i: \mu_i \neq 0} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\
&= \left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), 0) \\
&\quad + \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i).
\end{aligned}$$

Now we discuss the null part and non-null part separately.

(i) The risk can be rewritten as follows.

$$\begin{aligned}
& \left(\sum_{j \in S_{\text{TN}}} + \sum_{j \in S_{\text{FP}}} \right) r_S(t(w_j, \alpha_n k_n, q_n), 0) \\
&\leq 4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)) \\
&\quad + 4|S_{\text{FP}}| (t(c_n, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(c_n, \alpha_n k_n, q_n))^{-2}) \phi(t(c_n, \alpha_n k_n, q_n)).
\end{aligned} \tag{S1.11}$$

In the same way as (S1.6), we have

$$4n (t(1, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(1, \alpha_n k_n, q_n))^{-2}) \phi(t(1, \alpha_n k_n, q_n)) \leq C_2 k_n q_n,$$

for the same C_2 as in Lemma S.2. Applying the same technique as (S1.7), we have

$$\begin{aligned} & 4|S_{\text{FP}}| (t(c_n, \alpha_n k_n, q_n))^{-3} (1 + 1.5(t(c_n, \alpha_n k_n, q_n))^{-2}) \phi(t(c_n, \alpha_n k_n, q_n)) \\ & \leq C_4 |S_{\text{FP}}|, \end{aligned}$$

for the same constant C_4 as in Lemma S.2. Combining the results above, we have

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), 0) \leq C_2 k_n q_n + C_4 |S_{\text{FP}}|,$$

when n is large enough.

(ii) The risk corresponding to the non-null part can be rewritten as follows.

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq |S_{\text{TP}}| [1 + (t(c_n, \alpha_n k_n, q_n))^2] + |S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2] \\ & \leq C_1 |S_{\text{TP}}| + |S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2], \end{aligned}$$

for the same constant C_1 as in Lemma S.1. According to (S1.8), we obtain

$$|S_{\text{FN}}| [1 + (t(1, \alpha_n k_n, q_n))^2] \leq C_5 |S_{\text{FN}}| \log(n/k_n), \quad (\text{S1.12})$$

Finally, we obtain

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq C_1 |S_{\text{TP}}| + C_5 |S_{\text{FN}}| \log(n/k_n). \end{aligned}$$

Therefore, we have shown that

$$\begin{aligned} & \sum_{i=1}^n r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq C_2 k_n q_n + C_4 |S_{\text{FP}}| + C_1 |S_{\text{TP}}| + C_5 |S_{\text{FN}}| \log(n/k_n), \end{aligned}$$

when $k(\boldsymbol{\mu}) \geq 2\alpha_n k_n$. Similarly, the same result holds for the term $\sum_i r_S(t(w_i, \beta_n k_n, q_n), \mu_i)$, under this case.

Case 2: When $k(\boldsymbol{\mu}) \leq 2\alpha_n k_n$.

Similar to the discussion in Case 2 of the proof of Lemma S.1, we focus on the term $\sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i)$, and it can be similarly divided as follows.

$$\begin{aligned} & \sum_{i=1}^n r_S(t(w_i, 1, q_n), \mu_i) \\ & = \left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \\ & \quad + \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, 1, q_n), \mu_i). \end{aligned}$$

We also discuss the null part and non-null part separately.

(i) The same technique used in the proof of Lemma S.2 leads to

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \leq q_n + C_6 |S_{\text{FP}}|,$$

where C_6 is the same constant as in Lemma S.2.

(ii) Using the same notation, the risk corresponding to the non-null part can be rewritten

as

$$\begin{aligned}
& \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, 1, q_n), \mu_i) \\
& \leq |S_{\text{TP}}| [1 + (t(w_s, 1, q_n))^2] + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2] \\
& = C_3 |S_{\text{TP}}| \log \log n + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2],
\end{aligned}$$

for the same constant C_3 as in Lemma S.1. According to (S1.10), we obtain

$$\eta |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2] \leq 2 |S_{\text{FN}}| \log(2n/q_n).$$

Combining all the results completes the proof. □

S1.6 Proof of Theorem 1

Proof. The analysis follows the same arguments as in the proof of Lemma S.2, except for the terms involving the sets S_{FN} and S_{FP} . We again consider the following two cases.

Case 1: When $k(\mu) \geq 2\alpha_n k_n$.

In this case, the convergent rates of terms

$$\sum_{i \in S_{\text{FP}}} r_S(t(w_i, \alpha_n k_n, q_n), 0) \quad \text{and} \quad \sum_{i \in S_{\text{FN}}} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i)$$

will change compared with the proof of Lemma S.2. For the null part, the same technique used in (S1.7) leads to

$$\sum_{i \in S_{\text{FP}}} r_S(t(w_i, \alpha_n k_n, q_n), 0) \leq |S_{\text{OR}}| c^{-3} (1 + 1.5c^{-2}) \phi(c) = O(k_n \log n),$$

where $c = \tilde{\Phi}^{-1}((1 - \varepsilon)/2)$. Hence, the total risk of null part is

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), 0) \leq o(k_n) + O(k_n \log n) = O(k_n \log n).$$

For the non-null part, the same technique used in (S1.9) yields

$$\begin{aligned} & \sum_{i \in S_{\text{FN}}} r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq |S_{\text{FN}}| \left[1 + (t(1, \alpha_n k_n, q_n))^2 \right] = O(k_n \log(n/k_n)). \end{aligned}$$

Therefore, the total risk of non-null part is

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, \alpha_n k_n, q_n), \mu_i) \\ & \leq O(k_n \log n) + O(k_n \log(n/k_n)) = O(k_n \log n). \end{aligned}$$

Case 2: When $k(\boldsymbol{\mu}) \leq 2\alpha_n k_n$.

In this case the asymptotic rates of terms

$$\sum_{i \in S_{\text{FP}}} r_S(t(w_i, 1, q_n), 0) \quad \text{and} \quad \sum_{i \in S_{\text{FN}}} r_S(t(w_i, 1, q_n), \mu_i)$$

will change. In the same way, we have the following results. For the null part, we have

$$\begin{aligned} \sum_{i \in S_{\text{FP}}} r_S(t(w_i, 1, q_n), 0) & \leq |S_{\text{FP}}| \exp \left\{ -\frac{1}{2} [t(c_n, 1, q_n)]^2 \right\} \\ & \leq |S_{\text{OR}}| \exp(-c^2/2) \\ & = O(k_n \log n), \end{aligned}$$

where $c = \tilde{\Phi}^{-1}((1 - \varepsilon)/2)$. Therefore, we have

$$\left(\sum_{i \in S_{\text{TN}}} + \sum_{i \in S_{\text{FP}}} \right) r_S(t(w_i, 1, q_n), 0) \leq O(1) + O(k_n \log n) = O(k_n \log n).$$

For the non-null part, we obtain

$$\begin{aligned} & \left(\sum_{i \in S_{\text{TP}}} + \sum_{i \in S_{\text{FN}}} \right) r_S(t(w_i, 1, q_n), \mu_i) \\ & \leq k_n [1 + (t(c_n, 1, q_n))^2] + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2] \\ & = o(k_n \log(n/k_n)) + |S_{\text{FN}}| [1 + (t(1, 1, q_n))^2] \\ & \leq o(k_n \log(n/k_n)) + O(k_n \log n) \\ & = O(k_n \log n). \end{aligned}$$

Therefore, the total risk is

$$\mathbb{E}_{\mu} \|\hat{\mu} - \mu\|^2 \leq o(1) + O(k_n \log n) + O(k_n) = O(k_n \log n).$$

The proof is completed by combining results in both cases. □

S1.7 Proof of Theorem 2

We begin by introducing a lemma illustrating the property of BH procedure used in screening external datasets.

Lemma S.4. *Under Conditions 1 and Condition 2', the following claims hold as $n \rightarrow \infty$ for each $m \in [M]$:*

- $\mathbb{P}(S_{\gamma_m}^{(m)} \subseteq \hat{S}^{(m)}) \geq 1 - o(n^{1-b_3-C^2})$;
- $\mathbb{P}(|\hat{S}^{(m)} \setminus S^{(m)}| \leq 3k_n^{(m)}) \geq 1 - e^{-c(\log n)^2}$ for some constant $c > 0$.

Proof of Lemma S.4. Consider the first claim. For each $m \in [M]$, note that the threshold $t_{BH}^{(m)}$ satisfies

$$t_{BH}^{(m)} = \sup\{t \in [0, 1] : t/\hat{F}_n^{(m)}(t) \leq q'\},$$

where $\hat{F}_n^{(m)}(t) = n^{-1} \sum_{i=1}^n \mathbb{I}\{p_i^{(m)} \leq t\}$, $p_i^{(m)} = 2[1 - \Phi(|y_i^{(m)}|)]$. Since $\{p_i^{(m)} > (1 - \gamma_m)k_n^{(m)}q'/n\} \subseteq \{|y_i^{(m)}| < \Phi^{-1}(1 - (1 - \gamma_m)k_n^{(m)}q'/(2n))\}$ and $\Phi^{-1}(1 - (1 - \gamma_m)k_n^{(m)}q'/(2n)) = \left\{2 \log(2n/((1 - \gamma_m)k_n^{(m)}q'))\right\}^{1/2} - r$ for some relatively small $r \in [0, 1.5]$ by Lemma 12.3 of Abramovich et al. (2006), for $i \in S_{\gamma_m}^{(m)}$ we have

$$\begin{aligned} \mathbb{P}(p_i^{(m)} > (1 - \gamma_m)k_n^{(m)}q'/n) \\ &\leq \mathbb{P}(|\mu_i^{(m)}| - |\epsilon_i^{(m)}| < \Phi^{-1}(1 - (1 - \gamma_m)k_n^{(m)}q'/(2n))) \\ &\leq \mathbb{P}\left(|\epsilon_i^{(m)}| > \mu_{\gamma_m}^{(m)} - \sqrt{2 \log \frac{2n}{(1 - \gamma_m)k_n^{(m)}q'}}\right) \\ &\leq O((\log n)^{-1/2}) \exp\left\{-\frac{1}{2}C \left(\sqrt{2 \log n}\right)^2\right\}, \end{aligned}$$

where the last inequality holds due to the definition of $\mu_{\gamma_m}^{(m)}$. It leads to

$$\mathbb{P}(p_i^{(m)} > (1 - \gamma_m)k_n^{(m)}q'/n) = O\left((\log k_n^{(m)})^{-1/2}n^{-C^2}\right).$$

Therefore, we have

$$\begin{aligned} \mathbb{P}\left(\max_{i \in S_{\gamma_m}^{(m)}} p_i^{(m)} \geq \frac{k_n^{(m)}q'}{n}\right) &= O\left(k_n^{(m)}(\log k_n^{(m)})^{-1/2}n^{-C^2}\right) \\ &= o(n^{1-b_3-C^2}), \end{aligned}$$

where the first inequality holds due to Condition Condition 2'. In the meantime, under event

$$\left\{\max_{i \in S_{\gamma_m}^{(m)}} p_i^{(m)} \leq \frac{(1 - \gamma_m)k_n^{(m)}q'}{n}\right\},$$

the value $(1 - \gamma_m)k_n^{(m)}q'/n \in \{t \in [0, 1] : \hat{F}_n^{(m)}(t) \geq t/q'\}$, and hence $t_{BH}^{(m)} \geq (1 - \gamma_m)k_n^{(m)}q'/n$, i.e. the individuals in $S_{\gamma_m}^{(m)}$ are rejected by BH procedure. Therefore, the first claim holds.

The second claim holds according to Proposition S.1 and the claim that $k(\boldsymbol{\mu}^{(m)}) \leq 2k_n^{(m)}$ when n is large enough from Section 5 of Abramovich et al. (2006). We hence omit the proof. $\square$

Proof of Theorem 2. Let event $A_n = \{k_-(\boldsymbol{\mu}) \leq \ell \leq k_+(\boldsymbol{\mu})\}$, and define event $B_n = \{|S_{\text{FN}}| \leq C'k_n/(1 + \log(2n)/\log \log n)\} \cap \{|S_{\text{FP}}| \leq 3MC_Mk_n\}$, it suffices to show that

- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\} = o(1)$;
- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n^c\} = o(1)$;
- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n\} \leq \max\{C_2k_nq_n + (3MC_4C_M + C_1)k_n + C_5C'k_n \log \log n, q_n + 3MC_6C_Mk_n + (C_3 + 2C')k_n \log \log n\}$.

We first consider the probability of B_n . Note that $S_{\text{FN}} = S \setminus \hat{S} = \cap_{m=1}^M (S \setminus \hat{S}^{(m)})$. Since by Lemma S.4 we have

$$\mathbb{P}(A^M \subseteq \hat{S}) \geq \mathbb{P}(\cap_{m=1}^M \{S_{\gamma_m}^{(m)} \subseteq \hat{S}^{(m)}\}) \geq 1 - o(n^{1-b_3-C^2}). \quad (\text{S1.13})$$

According to (S1.13), we have

$$\mathbb{P}(|S \setminus \hat{S}| \leq |S \setminus A^M|) \geq 1 - o(n^{1-b_3-C^2}).$$

By Condition 3, it holds that

$$\mathbb{P}(|S \setminus \hat{S}| \leq C'k_n/(1 + \log(2n)/\log \log n)) \geq \mathbb{P}(|S \setminus \hat{S}| \leq |S \setminus A^M|) \geq 1 - o(n^{1-b_3-C^2}).$$

Then consider the probability of event $\{|S_{\text{FP}}| \leq C_Mk_n\}$. By the definition of S_{FP} , we have

$|S_{\text{FP}}| \leq |\hat{S} \setminus S| = \left| \bigcup_{m=1}^M (\hat{S}^{(m)} \setminus S) \right|$. According to Lemma S.4, we have

$$\mathbb{P} \left(\bigcup_{m=1}^M \{|\hat{S}^{(m)} \setminus S^{(m)}| > 3k_n^{(m)}\} \right) \leq M \exp \{-c(\log n)^2\},$$

which yields

$$\mathbb{P}(|S_{\text{FP}}| \leq 3MC_M k_n) \geq \mathbb{P}(\cap_{m=1}^M \{|\hat{S}^{(m)} \setminus S^{(m)}| \leq 3k_n^{(m)}\}) \geq 1 - e^{-c(\log n)^2}.$$

Combining the results above, we have

$$\mathbb{P}(B_n) \geq 1 - o(n^{1-b_3-C^2}) - e^{-c(\log n)^2}.$$

Now consider the three claims at the beginning of the proof. The first claim is proven in the proof of Lemma S.2. For the second claim, using the technique in (S1.1), we have

$$\mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n^c\} \leq O(n \log n)(o(n^{1-b_3-C^2}) + e^{-c(\log n)^2}) = o(1),$$

by the assumption that $C > \sqrt{2-b_3}$.

Finally, consider the third claim. According to the proof of Lemma S.2, in the event B_n we have

$$\begin{aligned} \mathbb{E} [\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} | \{\hat{w}_i\}_{i=1}^n] &\leq \max\{C_2 k_n q_n + (3MC_4 C_M + C_1)k_n + C_5 C' k_n \log \log n, \\ &\quad q_n + 3MC_6 C_M k_n + (C_3 + 2C')k_n \log \log n\}, \end{aligned}$$

where the constants are the same as those in Lemma S.2. Therefore, we have

$$\begin{aligned} \mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n\} &\leq \max\{C_2 k_n q_n + (3MC_4 C_M + C_1)k_n + C_5 C' k_n \log \log n, \\ &\quad q_n + 3MC_6 C_M k_n + (C_3 + 2C')k_n \log \log n\}, \end{aligned}$$

by the independence of weights $\hat{w}_i$'s and target dataset. Combining the results above completes the proof. $\square$

S1.8 Proof of Theorem 3

Proof. Let event $A_n = \{k_-(\boldsymbol{\mu}) \leq \ell \leq k_+(\boldsymbol{\mu})\}$ and define event $B_n = \{|S_{\text{FN}}| \leq \eta k_n\} \cap \{|S_{\text{FP}}| \leq 3MC_M k_n\}$, it suffices to show that

- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\} = o(1)$;
- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n^c\} = o(1)$;
- $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n\} \leq \max\{(C_2 q_n + 3MC_M C_4 + C_1)k_n + C_5 \eta k_n \log(n/k_n), q_n + 3MC_M C_6 k_n + C_3 k_n \log \log n + 8\eta k_n \log n\}$.

Note that in this case Lemma S.4 still holds. Analogous to the proof of Theorem 2, by Condition 4 and Lemma S.4, we have

$$\mathbb{P}(|S \setminus \hat{S}| \leq \eta k_n) \geq \mathbb{P}(|S \setminus \hat{S}| \leq |S \setminus A^M|) \geq 1 - o(n^{1-b_3-C^2}).$$

On the other hand, use the same technique in the proof of Theorem 2 we have $\mathbb{P}(|S_{\text{FP}}| \leq 3MC_M k_n) \geq 1 - e^{-c(\log n)^2}$. Combining the results above, we have

$$\mathbb{P}(B_n) \geq 1 - o(n^{1-b_3-C^2}) - e^{-c(\log n)^2}.$$

Next consider the three claims at the beginning of the proof. The first claim is proven in the proof of Lemma S.1. The second claim holds in the same reason as discussed in the proof of Theorem 2. Consider the third claim. According to the proof of Lemma S.3, in the

event B_n we have

$$\begin{aligned}
& \mathbb{E} [\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} | \{\hat{w}_i\}_{i=1}^n] \\
& \leq \max\{C_2 k_n q_n + C_4 |S_{\text{FP}}| + C_1 |S_{\text{TP}}| + C_5 |S_{\text{FN}}| \log(n/k_n), \\
& \quad q_n + C_6 |S_{\text{FP}}| + C_3 |S_{\text{TP}}| \log \log n + 8 |S_{\text{FN}}| \log n\} \\
& \leq \max\{C_2 k_n q_n + 3MC_M C_4 k_n + C_1 k_n + C_5 \eta k_n \log(n/k_n), \\
& \quad q_n + 3MC_M C_6 k_n + C_3 k_n \log \log n + 8\eta k_n \log n\},
\end{aligned}$$

where the constants are the same as those in Lemma S.2. Therefore, we have

$$\begin{aligned}
& \mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n\} \\
& = \mathbb{E} \left\{ \mathbb{I}\{B_n\} \mathbb{E} [\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n\} | \{\hat{w}_i\}_{i=1}^n] \right\} \\
& \leq \max\{(C_2 q_n + 3MC_M C_4 + C_1) k_n + C_5 \eta k_n \log(n/k_n), \\
& \quad q_n + 3MC_M C_6 k_n + C_3 k_n \log \log n + 8\eta k_n \log n\},
\end{aligned}$$

as the weights $\hat{w}_i$'s are independent of target dataset. Combining the results above completes the proof. $\square$

S1.9 Proof of Theorem 1'

Proof. Let event $A_n = \{k_-(\boldsymbol{\mu}) \leq \ell \leq k_+(\boldsymbol{\mu})\}$, and define event $B_n = \{|S_{\text{FP}}| = O(k_n \log n)\}$. It holds that $\mathbb{P}(|S_{\text{FP}}| = O(k_n \log n)) \geq 1 - e^{-c(\log n)^2}$ with the same technique in the proof of Theorem 2, which yields $\mathbb{P}(B_n) \geq 1 - e^{-c(\log n)^2}$. In the same way as the proof of Theorem 2, the following claims hold, and we omit the proof.

- $\mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n^c\} = o(1)$;
- $\mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n^c\} = o(1)$;
- $\mathbb{E} \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 \mathbb{I}\{A_n \cap B_n\} = O(k_n \log n)$.

Combining these claims completes the proof. $\square$

S2 Intuitions of weights construction

In this section we discuss the intuitions about the weights (2.7). The construction is motivated from adaptive Lasso (Zou, 2006), whose optimization function is as follows (the interesting vector is denoted as $\boldsymbol{\mu}$):

$$\arg \min_{\boldsymbol{\mu}} \|\mathbf{y} - \sum \mathbf{x}_j \mu_j\|^2 + \lambda \sum_{j=1}^p \hat{w}_j |\mu_j|,$$

where $\hat{\mathbf{w}} = 1/|\hat{\boldsymbol{\mu}}|^\gamma$ for some initial estimate $\hat{\boldsymbol{\mu}}$ and some constant γ . In this scheme, a large initial estimate for μ_i reduces the effective threshold, suggesting that index i is likely a signal; conversely, a small initial estimate increases the threshold, indicating a likely null.

Motivated by this analogy, we reinterpret the original threshold t_F —used in the BH procedure—as the counterpart to λ in the Lasso formulation. We want to find the similarity between $\tilde{\Phi}^{-1}(\hat{w}_i x)/\tilde{\Phi}^{-1}(x)$ with $1/|\hat{\mu}_i^{\text{initial}}|^\gamma$. To this end, we express the weighted threshold $t(\hat{w}_i, \ell, q)$ in a form analogous to the adaptive Lasso ($\lambda/|\hat{\mu}_i^{\text{initial}}|^\gamma$) as

$$u(\hat{w}_i, x) := \tilde{\Phi}^{-1}(\hat{w}_i x)/\tilde{\Phi}^{-1}(x),$$

where we define $x = \ell q/(2n)$ for simplicity. For small x and $\hat{w}_i x \rightarrow 0$, we can approximate the scaling function $u(\hat{w}_i, x) \approx 1 - \frac{\log \hat{w}_i}{\log(\frac{2n}{\ell q})}$. Comparing this with the adaptive Lasso penalty factor $1/|\hat{\mu}_i^{\text{initial}}|^\gamma$, we observe that

$$1 - \frac{\log \hat{w}_i}{\log\left(\frac{2n}{\ell q}\right)} = \frac{1}{|\hat{\mu}_i^{\text{initial}}|^\gamma} \quad \Rightarrow \quad \log \hat{w}_i = \left(1 - \frac{1}{|\hat{\mu}_i^{\text{initial}}|^\gamma}\right) \log\left(\frac{2n}{\ell q}\right).$$

For signal indices $i \in S$, where the initial estimates $\hat{\mu}_i^{\text{initial}}$ tend to be large, the above expression suggests that $\log \hat{w}_i = O(\log \frac{n}{\ell q})$. Taking into account the scaling of ℓ , $|S|$, and q ,

where ℓ and q are always much smaller than n , this leads to the approximation

$$\hat{w}_i \approx n^\rho \tag{S2.1}$$

for some constant $\rho > 0$. Conversely, for null indices with small initial estimates, we expect $\log \hat{w}_i \rightarrow -\infty$, implying $\hat{w}_i < 1$ and tending toward zero. Based on these insights, we choose the weights as follows: for indices in $\hat{S}$, we set $\hat{w}_i$ to be $n/\log n$, which aligns with (S2.1) while maintaining the condition $\hat{w}_i \ell q/n \rightarrow 0$. For indices not in $\hat{S}$, we set $\hat{w}_i = 1$, since the threshold t_F already adapts to the global sparsity level k_n of the target parameter (Abramovich et al., 2006) and effectively filters out non-signals.

This design markedly differs from the adaptive Lasso, which relies on a tuning parameter λ and does not exploit information about global sparsity. In this way, the weights that encode unit-level support information are seamlessly integrated with the global sparsity adaptation through (2.6), highlighting the novelty of weighting strategy in SATL.

S3 Additional Numerical Results

S3.1 Additional Simulation Studies

In this section we present two additional simulation studies under the same model in Section 4.

Experiment 1. We investigate how the number of informative external datasets influences performance in this setting. When determining the size of overlap between $\boldsymbol{\mu}^{(m)}$ and $\boldsymbol{\mu}$, we randomly sample M_I indices from $[M]$ and assign their similarity ratio to be 0.3, while the similarity ratios for the remaining external datasets are set to zero. The remaining steps for generating the supports of the source data are the same as described above. We compare the performance of different methods under varying d and numbers of informative external

datasets M_I as follows

$$d \in \{0.5, 7, 13.5, 20\}, \quad M_I \in \{0, 1, 2, 3, 5, 7, 9, 10\}.$$

The performance of all methods under these settings is summarized in Figure S.1. As expected, the performance of ABDJ remains unchanged across all scenarios. The performance of all methods improves as the number of informative external datasets increases, whereas SATL consistently outperforms the competing methods. When the number of informative external datasets decreases, the risks of existing transfer learning methods grow, whereas SATL is less affected. This robustness arises because the remaining informative datasets can still be integrated to provide useful support information, consistent with Theorem 2. Specifically, SATL achieves low MSE even when the number of informative external datasets is not large (e.g., $M_I \in [3, 5]$), demonstrating that structure transfer is highly flexible. When $M_I \in [0, 3]$, the performance of SATL is close to ABDJ, corresponding to the large- γ scenario in Theorem 3. By contrast, existing transfer learning methods are more heavily influenced when there are more adverse external datasets. Overall, SATL exhibit both flexibility in weakly informative scenarios and robustness against adverse external information.

Experiment 2. In this setting, we investigate the performance of methods under different signal strengths to evaluate robustness. The parameters follow the same setting as in Section 4, except that the target parameter is specified as follows: $(\mu_1, \dots, \mu_{k_n})$ are evenly spaced between 0.1 and $\mu_{\max}$, and $\mu_i = 0$ for all $i > k_n$. The similarity ratios are fixed at $r_m = 0.3$ for all $m \in [M]$. We compare the performance of different methods across varying d and signal strength $\mu_{\max}$:

$$d \in \{0.5, 7, 13.5, 20\}, \quad \mu_{\max} \in \{2, 5.6, 9.2, 12.8, 16.4, 20\}.$$

The results are summarized in Figure S.2. As expected, the performance of the ABDJ

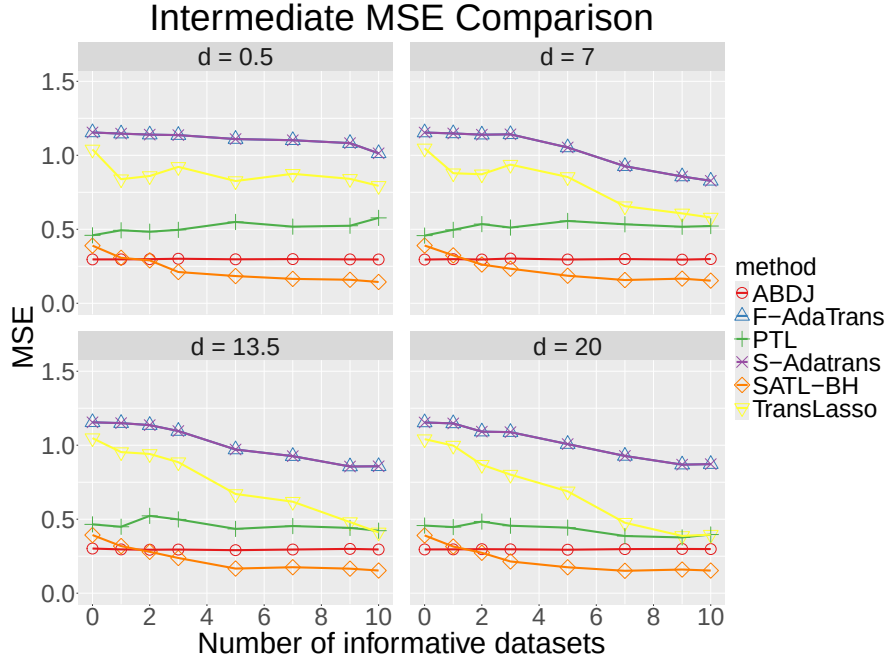


Figure S.1: Estimation performance of different methods in Experiment 1.

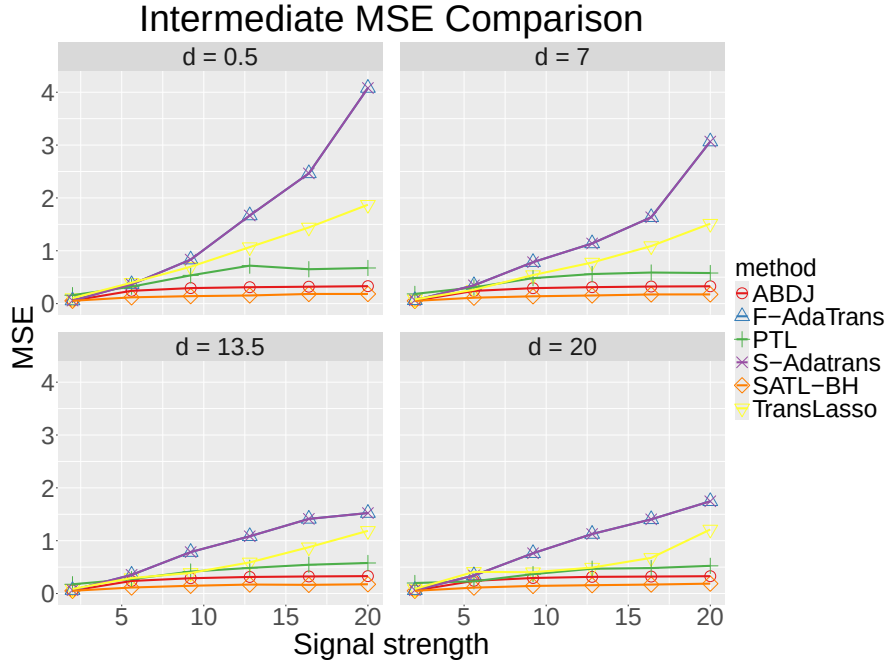


Figure S.2: Estimation performance of different methods in Experiment 2.

method remains unaffected by changes in d . SATL achieves lower estimation errors compared to both transfer learning methods and the ABDJ method, demonstrating robust performance across different values of d and varying sparsity levels.

S3.2 Application to Covariate-Assisted Inference

In this section, we present a performance comparison of different methods for high-dimensional estimation with covariate assistance, motivated by [Cai et al. \(2019\)](#). We follow the model considered in [Cai et al. \(2019\)](#). Suppose that $\{X_{ij} : 1 \leq j \leq n_x\}$ and $\{Y_{ik} : 1 \leq k \leq n_y\}$, $i = 1, \dots, n$, are repeated measurements of independent random variables X_i and Y_i respectively, obeying:

$$X_{ij} = \mu_{xi}^* + \epsilon_{xij},$$

$$Y_{ik} = \mu_{yk}^* + \epsilon_{yik},$$

where $\epsilon_{xij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{xi}^2)$ and $\epsilon_{yik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{yi}^2)$. Let $N = n_x + n_y$. Denote $\gamma_x = n_x/N$ and $\gamma_y = n_y/N$. Consider sample means $\bar{X}_i = n_x^{-1} \sum_{j=1}^{n_x} X_{ij}$ and $\bar{Y}_i = n_y^{-1} \sum_{k=1}^{n_y} Y_{ik}$. The target and external datasets are denoted as $\mathbf{T}_1$ and $\mathbf{T}_2$, calculated as below:

$$(T_{1i}, T_{2i}) = \sqrt{\frac{n_x n_y}{2N}} \left(\frac{\bar{X}_i - \bar{Y}_i}{\sigma_{pi}}, \frac{\bar{X}_i + \kappa^* \bar{Y}_i}{\sqrt{\kappa^*} \sigma_{pi}} \right),$$

where $\sigma_{pi}^2 = \gamma_y \sigma_{xi}^2 + \gamma_x \sigma_{yi}^2$. There is an additional $\sqrt{1/2}$ compared to original statistics in [Cai et al. \(2019\)](#) in order to make the variance be one. We fix $n_x = n_y = 50$, $\sigma_{xi} = 1$ and $\sigma_{yi} = 2$. Our simulations use $n = 2000$. We consider the following two cases, where different methods are applied to simulated data and the results are averaged over 200 replications.

- *Setting 1:* we set $\mu_{x,1:k} = 12$, $\mu_{x,(k+1):(2k)} = 6$, $\mu_{x,(2k+1):n} = 0$, $\mu_{y,1:k} = 10$, $\mu_{y,(k+1):(2k)} = 6$ and $\mu_{y,(2k+1):n} = 0$. We vary k from 50 to 450 to investigate the effect of sparsity.
- *Setting 2:* we use k_1 and k_2 to denote the number of non-zero coordinates and the number of coordinates with differential effects respectively. Let $\mu_{x,1:k_2} = 12$, $\mu_{x,(k_2+1):k_1} = 6$, $\mu_{x,(k_1+1):n} = 0$, $\mu_{y,1:k_2} = 10$, $\mu_{y,(k_2+1):k_1} = 6$ and $\mu_{y,(k_1+1):n} = 0$. We fix $k_1 = 1000$ and vary k_2 from 50 to 450. This setting investigates how the informativeness of the auxiliary datasets would affect the performance of different methods.

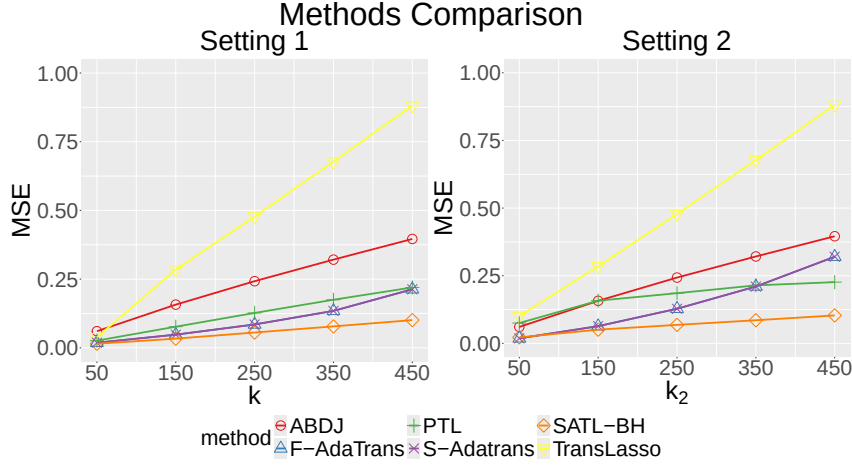


Figure S.3: Estimation performance of different methods.

The estimation performance is plotted as functions of various parameter values and displayed in Fig. S.3. Our method demonstrates higher efficiency compared with conventional transfer learning methods and ABDJ. Since the external dataset $\mathbf{T}_2$ shares little information about signal values, standard transfer learning methods fail to utilize it effectively sufficiently. In contrast, our method exploits the support information in $\mathbf{T}_2$, as $\text{supp}(\mathbf{T}_2)$ adequately covers $\text{supp}(\mathbf{T}_1)$, resulting in superior estimation performance.

S4 Comparison with Group Lasso

In this section, we investigate the empirical performance of SATL and compare it with Group Lasso methods (Yuan and Lin, 2006), which are widely used in multi-task learning problems (Lounici et al., 2011) and are closely related to the transfer learning setting considered in this paper.

Both transfer learning and multi-task learning aim to leverage information across related tasks. However, the underlying structural assumptions differ substantially. In our framework, the external datasets are allowed to share only partial support overlap with the target parameter. The intersection between the support of the target parameter and the supports of the external parameters may even be empty. By contrast, traditional Group Lasso methods typically assume that multiple tasks share a common sparsity structure, induced

through a sparsity penalty that encourages identical or highly similar supports across tasks (Lounici et al., 2011; Knight and Duan, 2023). Consequently, when the external datasets contain heterogeneous supports that differ substantially from the target support, which is common in transfer learning applications, the structural assumption underlying Group Lasso may be violated. In such settings, the shared support identified by Group Lasso may only capture a small subset of the true target support, potentially leading to inferior estimation performance. The following simulation study illustrates this phenomenon.

To facilitate a meaningful comparison across methods, we consider the identity design matrix when implementing the transfer learning procedures. The target and external datasets are generated independently from model (2.2) with unit variances. We fix the sample size $n = 200$, target sparsity level $k_n = 20$, and number of external datasets $M = 6$. The target parameter is generated as follows: the first k_n nonzero coordinates $(\mu_1, \dots, \mu_{k_n})$ are evenly spaced between 0.1 and 5, while the remaining coordinates are set to zero. For the external datasets, we specify the support sizes as $(k^{(1)}, \dots, k^{(M)}) = (10, 16, 22, 28, 34, 40)$. The overlap between the support S of the target parameter and the support $S^{(m)}$ of the m -th external parameter is controlled through

$$|S \cap S^{(m)}| = \min \{ \lceil r_m k_n \rceil, k^{(m)} \},$$

where $r_m \in (0, 1)$ denotes the *similarity ratio* introduced in Section 4. The support construction follows the same procedure as in Section 4.

Conditional on the generated supports, the nonzero coordinates of the external parameters are assigned as follows: (i) For coordinates in $S \cap S^{(m)}$, we generate $\mu_i^{(m)} = \mu_i + N(d, 0.1^2)$. (ii) For coordinates in $S^{(m)} \setminus S$, the signals are evenly spaced between 0.1 and 10. Once generated, the external parameters $\{\boldsymbol{\mu}^{(m)}\}_{m=1}^M$ remain fixed across all simulation replications.

We compare SATL and Group Lasso under varying levels of signal discrepancy d and

similarity ratio r_m . Specifically, we set $r_m = r$ for all m and consider

$$d \in \{0.5, 3, 5.5, 8\}, \quad r \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.4, 0.6, 0.8, 0.85, 0.9, 0.95\}.$$

The performance is evaluated using the average estimation error $\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2/n$. Each setting is replicated over 200 independent trials.

Figure S.4 summarizes the performance of the competing methods, where the horizontal axis represents the dissimilarity ratio $1 - r$. SATL demonstrates substantial efficiency gains across most settings, particularly when the external datasets are moderately informative yet exhibit heterogeneous supports. This robustness is consistent with the theoretical results established in Theorems 2 and 3. In contrast, group lasso performs consistently worse than SATL in these settings. This is because one of the external datasets has a very sparse support, so that the supports of the external and target parameters may ultimately fail to share a common subset, limiting the effectiveness of approaches that rely heavily on shared support structures.

Overall, these results demonstrate that SATL is considerably more flexible and robust than Group Lasso in transfer learning scenarios involving heterogeneous auxiliary datasets.

S5 Extension to High-Dimensional Regression Models

In this section, we briefly discuss how the proposed SATL framework may be extended beyond the Gaussian sequence model considered in the main text. Although the compound estimation problem studied in this paper differs conceptually from standard regression problems, the key idea underlying SATL can still be adapted to more general high-dimensional inference settings.

Specifically, our framework is fundamentally designed for large-scale estimation problems in which one constructs many Gaussian summary statistics and then performs simultaneous estimation. In regression problems, by contrast, the observations are sample pairs (X_i, y_i)

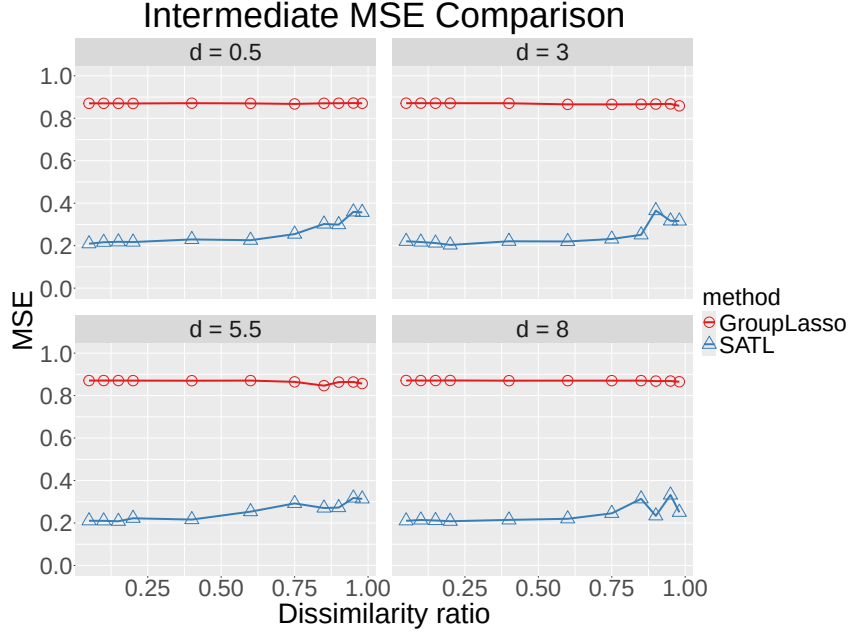


Figure S.4: Estimation performance of different methods.

and the primary goal is estimation of regression coefficients through the covariate-response relationship. Consequently, while sparse regression models can formally be related to Gaussian sequence models through asymptotic approximations, the statistical structures and methodological objectives are different. The extensions considered here should therefore be viewed as plug-in adaptations of SATL rather than direct applications of the original framework.

S5.1 Plug-in SATL via Debiased Estimators

Consider the high-dimensional linear regression model

$$y = X\beta + \varepsilon, \quad (\text{S5.1})$$

where $X \in \mathbb{R}^{n \times p_n}$ is the design matrix, $\beta \in \mathbb{R}^{p_n}$ is a sparse regression coefficient vector, and ε is random noise. In the high-dimensional regime $p_n \gg n$, the Lasso estimator is biased due to regularization. To obtain asymptotically normal estimators, we employ the debiased Lasso estimator (Van de Geer et al., 2014; Javanmard and Montanari, 2014), which admits

the approximate representation

$$\hat{\beta}_j^{db} = \beta_j + \frac{1}{\sqrt{n}}Z_j + o_p(n^{-1/2}), \quad (\text{S5.2})$$

where Z_j is asymptotically Gaussian under suitable regularity conditions. After standardization, the coordinates approximately follow a Gaussian sequence model, allowing SATL to be applied in a plug-in fashion.

More precisely, suppose we observe one target dataset and M external datasets. For each dataset, we first compute debiased estimators and their associated standardized statistics. The external statistics are then used to construct weighted support information through the screening step in SATL procedure. Finally, the target estimator is obtained by applying the shrinkage rule of SATL to the target debiased estimator coordinate-wise.

The same construction can also be extended to generalized linear models. For example, in high-dimensional logistic regression, debiased estimators similarly satisfy asymptotic normality under suitable assumptions, yielding approximately Gaussian standardized statistics. This enables the same weighted multiple-testing mechanism to be incorporated into logistic regression problems.

We emphasize that these extensions are currently conceptual and rely on approximate Gaussianity of debiased estimators. Additional technical challenges may arise from dependence across coordinates, estimation error in nuisance quantities, and finite-sample deviations from asymptotic normality. A rigorous theoretical analysis of these extensions is beyond the scope of the present paper and is left for future work.

S5.2 Simulation Setup: Linear Regression

We first investigate the extension of SATL to sparse high-dimensional linear regression. The target model is generated from (S5.1), where $\varepsilon \sim N(0, 1)$ and the rows of the design matrix are generated independently from a multivariate normal distribution $X_i \stackrel{\text{i.i.d.}}{\sim} N(0, \Sigma)$,

$i = 1, \dots, n$, where the covariance matrix $\Sigma = (\Sigma_{jk})_{1 \leq j, k \leq p_n}$ satisfies $\Sigma_{jk} = \rho^{|j-k|}$, with $\rho = 0.5$. We set the sample size and ambient dimension to $n = 200$, $p_n = 400$. The target sparsity level is fixed at $s = 40$, and the number of external datasets is $M = 6$.

The target regression coefficient vector β is generated as follows: the first 40 coordinates are evenly spaced between 0.1 and 10, while the remaining coordinates are set to zero. For the external datasets, the support sizes are specified as $(k^{(1)}, \dots, k^{(M)}) = (30, 38, 46, 54, 62, 80)$. To control the structural similarity between the target and external datasets, we define a similarity ratio $r \in (0, 1)$ following Section 4, and enforce $|S \cap S^{(m)}| = \min\{\lceil rk_n \rceil, k^{(m)}\}$.

Given the generated supports, the nonzero coefficients of the external datasets are assigned as follows: (i) For coordinates in $S \cap S^{(m)}$, we generate $\beta_j^{(m)} = \beta_j + N(d, 0.1^2)$, where d controls the magnitude difference between target and external signals. (ii) For coordinates in $S^{(m)} \setminus S$, the coefficients are evenly spaced between 0.1 and 20.

We compare the following methods:

- Lasso: standard Lasso estimator using only the target dataset;
- SATL: the proposed plug-in SATL estimator based on debiased statistics;
- GLMTrans: the transfer learning regression method proposed in Li et al. (2021).

We fix the SATL tuning parameters as $q = q' = 0.1$, $\varepsilon = 0.01$, $p = 1/M$. The simulations are conducted under $d \in \{0.5, 3.67, 6.83, 10\}$, and $r \in \{0.05, 0.1, 0.15, 0.25, 0.5, 0.8\}$. Each setting is replicated over 200 independent trials. The estimation performance is evaluated using the mean squared error $\|\hat{\beta} - \beta\|^2/p_n$.

Figure ?? summarizes the simulation results. The plug-in SATL estimator consistently improves upon both the standard Lasso and the debiased estimator, particularly when the external datasets share moderate support overlap with the target parameter. Moreover, SATL remains relatively robust when the signal magnitude difference d becomes large. This behavior is consistent with the main intuition of the proposed framework: SATL primar-

ily exploits transferable support information rather than requiring the external regression coefficients themselves to be numerically close to the target parameter.

By contrast, methods relying more heavily on coefficient similarity may deteriorate when the external datasets become heterogeneous. When the similarity ratio r becomes very small, the advantage of transfer learning diminishes and the performance of SATL gradually approaches that of the target-only estimators.

S5.3 Simulation Setup: Logistic Regression

We next consider sparse high-dimensional logistic regression. The response variables are generated according to

$$\Pr(y_i = 1 \mid X_i) = \frac{\exp(X_i^\top \beta)}{1 + \exp(X_i^\top \beta)}.$$

The target and external coefficient vectors and design matrixes are generated using the same support construction as in the linear regression setting. The sample size, ambient dimension, sparsity level, number of external datasets, and similarity ratios are fixed as before: $n = 200$, $p_n = 400$, $s = 40$, $M = 6$. We consider $d \in \{0.2, 1.8, 3.4, 5\}$, and $r \in \{0.05, 0.1, 0.15, 0.25, 0.5, 0.8\}$.

The following methods are compared:

- Logistic-Lasso: ℓ_1 -penalized logistic regression;
- SATL-Logistic: plug-in SATL based on debiased logistic statistics;
- GLMTrans: transfer learning logistic regression estimator (Tian and Feng, 2023).

The results are presented in Figure ???. The behavior is similar to the linear regression setting. In particular, the proposed SATL extension achieves stable gains over target-only methods and exhibits improved robustness under heterogeneous external datasets. These numerical results suggest that the main idea of SATL may remain effective beyond the Gaussian sequence model whenever approximately Gaussian debiased statistics are available.

S6 Robustness to Unknown Variances and Non-Gaussian Sampling Distributions

In this section, we investigate the empirical robustness of SATL beyond the model (2.2) with known unit variances. Since each Y_i is a summary statistics constructed from replicated observations, we consider more general settings in which the underlying observations may have unknown variances or non-Gaussian sampling distributions.

Throughout this section, we follow the same transfer learning setting as in Section 4. Specifically, we fix $n = 500$, target sparsity level $k_n = 40$, and number of external datasets $M = 8$. The target signal vector $\boldsymbol{\mu}$ is generated in the same manner as Section 4, with the first 40 coordinates evenly spaced between 0.1 and 10 and the remaining coordinates equal to zero. The support sizes of the external datasets are specified as $(k^{(1)}, \dots, k^{(M)}) = (30, 36, 42, 47, 53, 58, 64, 69, 75, 80)$. The overlap between the target and external supports is controlled through the similarity ratio $r \in \{0.05, 0.1, 0.15, 0.25, 0.5, 0.8\}$, and the signal magnitude difference parameter is chosen from $d \in \{0.5, 7, 13.5, 20\}$. The SATL tuning parameters are fixed at $q = q' = 0.1$, $\varepsilon = 0.01$, $p = 1/M$. Each setting is replicated over 200 independent trials. We compare the methods introduced in Section 4. The performance is evaluated using the average estimation error $\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2/n$.

S6.1 Unknown Variance Setting

We first consider the setting where the variances are unknown and must be estimated from replicated observations. For each unit $i \in [n]$, we observe

$$Y_{ij} = \mu_i + \sigma_i \epsilon_{ij}, \quad j = 1, \dots, n_i,$$

where $\epsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$, and the variance σ_i^2 is unknown and allowed to vary across coordinates. The external datasets are generated analogously with their own heterogeneous variances. In

the simulations, we fix the number of replicates at $n_i = 30$ for all $i \in [n]$, and generate the coordinate-wise standard deviations independently from the uniform distribution on $[0.5, 2]$. For each coordinate, we estimate the variance using the sample standard deviation

$$\hat{\sigma}_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2,$$

where

$$\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}.$$

We then construct the standardized statistics $Z_i = \frac{\sqrt{n_i} \bar{Y}_i}{\hat{\sigma}_i}$. Under moderate replication sizes, the standardized statistics are approximately Gaussian with unit variance. We therefore apply the SATL procedure to the normalized observations and transform the resulting estimators back to the original scale afterward. The competing methods are implemented analogously on the standardized scale. Figure S.5 summarizes the simulation results. Overall, SATL continues to substantially outperform the target-only estimator ABDJ and remains competitive compared with existing transfer learning methods. In particular, SATL remains robust when the similarity ratio is moderate or large. These results suggest that the SATL framework is reasonably robust to departures from the idealized assumption of known unit variances and can still effectively exploit transferable support information under heterogeneous variance settings.

S6.2 Non-Gaussian Sampling Distributions

We next investigate the robustness of SATL under non-Gaussian sampling distributions. For each coordinate $i \in [n]$, the replicated observations are generated according to

$$Y_{ij} = \mu_i + \epsilon_{ij}, \quad j = 1, \dots, n_i,$$

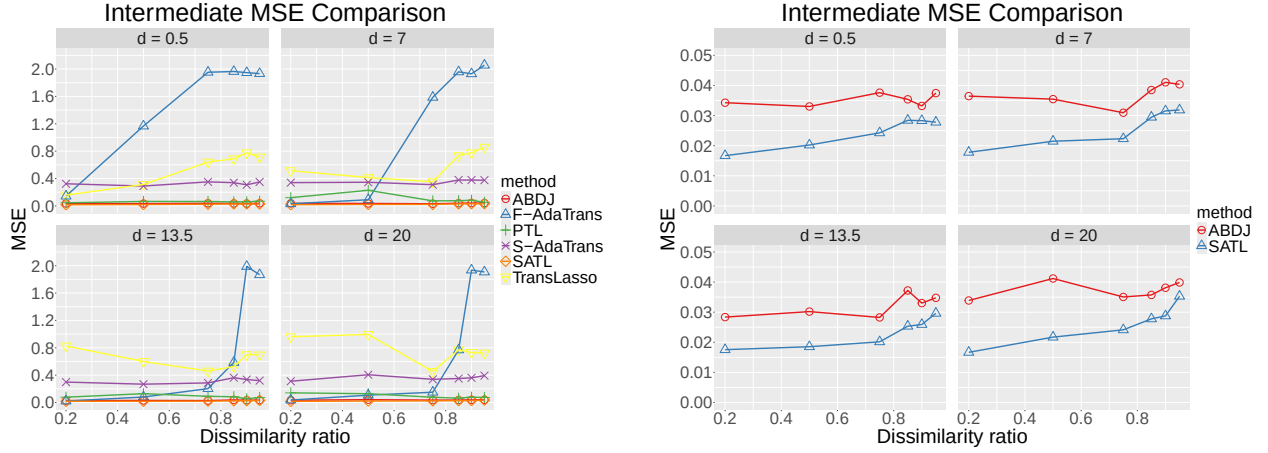


Figure S.5: Performance under unknown variances. Left: complete comparison. Right: zoomed-in portion to better resolve ABDJ and SATL, which are difficult to distinguish in the left figure.

where the noise variables ϵ_{ij} are independently generated from centered uniform distribution: $\epsilon_{ij} \sim \text{Uniform}(-\sqrt{3}, \sqrt{3})$. The number of replicates is again fixed at $n_i = 30$. For each coordinate, we construct the standardized statistic $Z_i = \frac{\sqrt{n_i} \bar{Y}_i}{S_i}$, where $\bar{Y}_i$ and S_i denote the sample mean and sample standard deviation, respectively. Due to standard central limit theorem arguments, the resulting standardized statistics are approximately Gaussian. To assess the robustness of the original methodology under model misspecification, we directly apply the Gaussian-calibrated SATL procedure developed in Section 2, using the same normal quantile thresholds without further modification.

The simulation result is presented in Figure S.6, respectively. Overall, SATL continues to exhibit stable and competitive performance. In particular, SATL consistently improves upon the target-only estimator and remains relatively robust across different similarity ratios. The risk gap between SATL and ABDJ is less pronounced than in the unknown-variance case, primarily because the uniform noise yields summary statistics that are less Gaussian when incorporated into SATL. Nonetheless, these observations indicate the potential applicability of SATL to a broader range of settings.

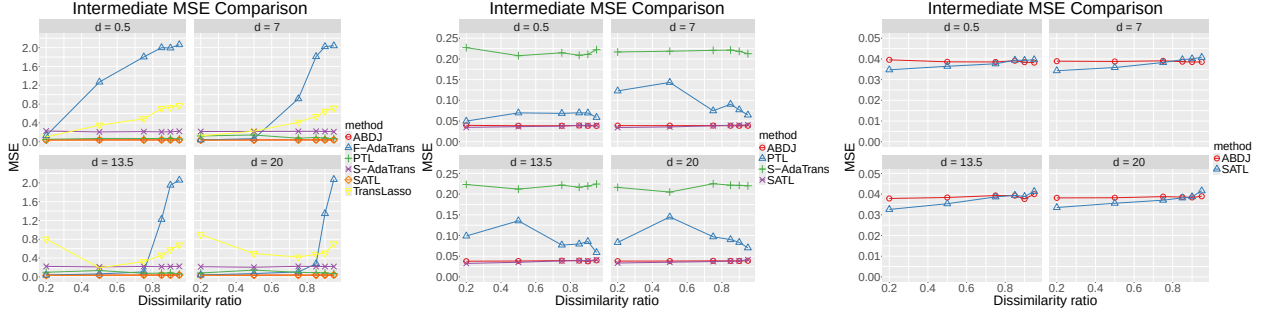


Figure S.6: Performance under non-Gaussian noise. Left: complete comparison; Middle: zoomed-in portion to better resolve ABDJ, SATL, PTL, and S-AdaTrans; Right: zoomed-in portion to better resolve ABDJ and SATL.

S7 Sensitivity Analysis for Hyperparameters in SATL

In this section, we investigate the empirical sensitivity of SATL with respect to the three hyper-parameters involved in the procedure: the target FDR level q , the external screening FDR level q' , and the aggregation parameter p . The goal of this study is to complement the theoretical discussion in Section 3 and to provide practical guidance on parameter selection.

Throughout this section, we consider the same model (2.2) and simulation framework as in Section 4. Specifically, we set $n = 1000$, $k_n = 40$, and $M = 10$, and generate the target signals with nonzero coordinates equally spaced between 0.1 and 10. For the external datasets, the supports are generated in the same manner as in Section 4, and the similarity ratio between the target and external supports varies over $r \in \{0.05, 0.1, 0.15, 0.25, 0.5, 0.8\}$.

Given the generated supports of the external parameters, the true parameters of the m -th source dataset are assigned in the same way as in Section 4, while the parameter d is fixed at 7. For each configuration, the results are averaged over 200 replications.

S7.1 Sensitivity to the target FDR level q

We first study the sensitivity of SATL to the target FDR level q , which determines the adaptive threshold in the final estimation step. In this experiment, we fix $q' = 0.1$, $p = \frac{1}{M}$, and vary $q \in \{0.01, 0.03, \dots, 0.29\}$.

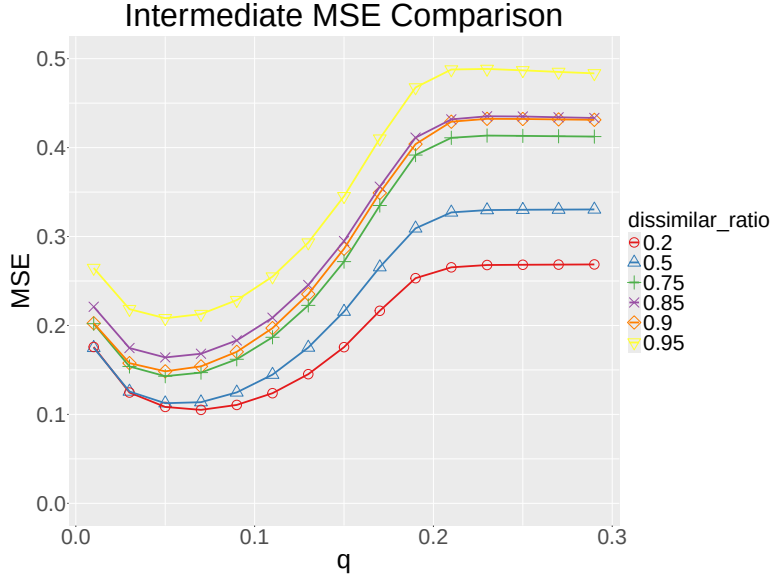


Figure S.7: SATL Performance under different q .

Figure S.7 reports the resulting MSE as a function of q under different similarity ratios. Overall, the performance of SATL is relatively stable across a broad range of q values. When q is extremely small, the procedure becomes more conservative, leading to larger thresholds and slightly increased bias. As q increases, the thresholds become less conservative and the MSE initially decreases. For very large values of q , the procedure may include more noise coordinates, resulting in a moderate increase in risk. Nevertheless, the overall variation in performance remains limited, indicating that SATL is not highly sensitive to the precise choice of q . These observations are consistent with the theoretical discussion in Section 3. The asymptotic condition $q \rightarrow 0$ is mainly a technical requirement used in the risk analysis, while in finite samples a fixed moderate value such as $q = 0.1$ provides a good balance between adaptivity and stability.

S7.2 Sensitivity to the screening FDR level q'

Next, we investigate the effect of the screening FDR level q' , which is used in the BH screening step for the external datasets. In this experiment, we fix $q = 0.1$, $p = \frac{1}{M}$, and vary $q' \in \{0.01, 0.03, \dots, 0.49\}$. The results are shown in Figure S.8. The influence of q' on the

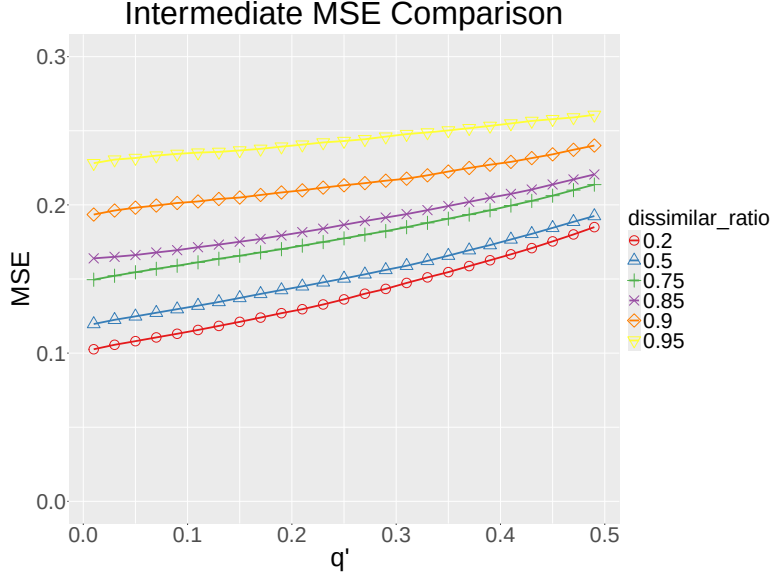


Figure S.8: SATL Performance under different q' .

final MSE is weak. This behavior can be explained by the role of q' in the algorithm: its primary purpose is to control the number of false positives in the estimated external supports $\hat{S}^{(m)}$. As long as q' remains within a reasonable range, the resulting aggregated support set $\hat{S}$ remains stable, and the corresponding weights used in the transfer step are only mildly affected, consistent with the discussion of q' in Section 3.

S7.3 Sensitivity to the aggregation parameter p

Finally, we study the aggregation parameter p , which determines how many external datasets must simultaneously select a coordinate before it is included in the aggregated support $\hat{S}$. We fix $q = q' = 0.1$, and vary $p \in \{\frac{1}{M}, \frac{2}{M}, \dots, \frac{9}{M}\}$.

The corresponding results are displayed in Figure S.9. Analogous to the roles of q and q' , the parameter p also has a relatively limited impact on the transfer performance. Smaller values of p correspond to a more aggressive aggregation strategy, allowing information from any external dataset to contribute to the candidate support. This increases the amount of transferred information and generally improves performance, especially when the external datasets share only limited overlap with the target support. As p increases, the aggregation

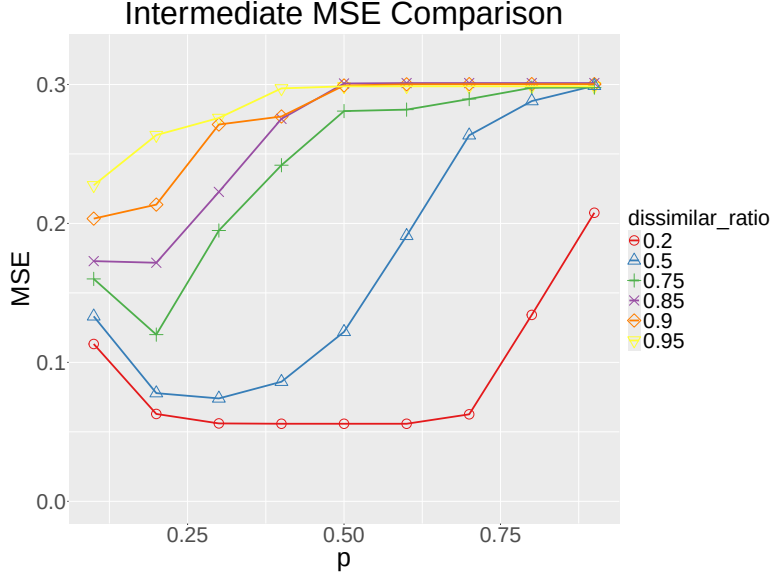


Figure S.9: SATL Performance under different q' .

becomes more conservative, since stronger agreement across external datasets is required. This reduces the influence of potentially misleading external information and can improve robustness under cases when external datasets have high similarity ratios. However, when the external datasets are only weakly informative and exhibit substantial heterogeneity, overly large values of p may exclude useful information and hinder effective aggregation of the target support information. Consequently, the transfer effect may weaken, leading to a larger estimation error.

Overall, the results suggest that the default choice $p = \frac{1}{M}$ provides a favorable balance between transfer efficiency and robustness in the considered settings. At the same time, choosing a moderately larger value of p may be beneficial when it is known that most external datasets provide reliable information about the target support.

The sensitivity experiments demonstrate that SATL is empirically robust to the choices of the hyper-parameters q , q' , and p . In particular, the method exhibits stable performance over broad ranges of q and q' , while the aggregation parameter p provides a natural mechanism for balancing transfer efficiency and robustness against adverse auxiliary information. These findings support the practical use of simple default choices such as $q = q' = 0.1$, $p = \frac{1}{M}$,

Table S.1: Summary of RNA-seq data

Dataset	Time span	Organ	Tackle	Replicates
Acute_Colon_Control	6 hours	colon	water	3
Acute_Colon_Imeglimin	6 hours	colon	Imeglimin	3
Acute_Colon_Metformin	6 hours	colon	Metformin	3
Chronic_ProximalSI_Control	12 weeks	proximal small intestine	water	3
Chronic_ProximalSI_Imeglimin	12 weeks	proximal small intestine	Imeglimin	3
Chronic_ProximalSI_Metformin	12 weeks	proximal small intestine	Metformin	3
Chronic_MidleSI_Control	12 weeks	middle small intestine	water	3
Chronic_MidleSI_Imeglimin	12 weeks	middle small intestine	Imeglimin	3
Chronic_MidleSI_Metformin	12 weeks	middle small intestine	Metformin	3
Chronic_DistalSI_Control	12 weeks	distal small intestine	water	3
Chronic_DistalSI_Imeglimin	12 weeks	distal small intestine	Imeglimin	3
Chronic_DistalSI_Metformin	12 weeks	distal small intestine	Metformin	3
Chronic_Colon_Control	12 weeks	colon	water	3
Chronic_Colon_Imeglimin	12 weeks	colon	Imeglimin	3
Chronic_Colon_Metformin	12 weeks	colon	Metformin	3

which were adopted throughout the numerical experiments in this paper.

S8 Additional information of real data analysis

In this section we present the heatmaps of all datasets and the summary of experimental conditions and organs of datasets in Section 5. The data can be downloaded at

www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE304061.

Table S.1 presents the conditions for all groups of RNA-sequence.

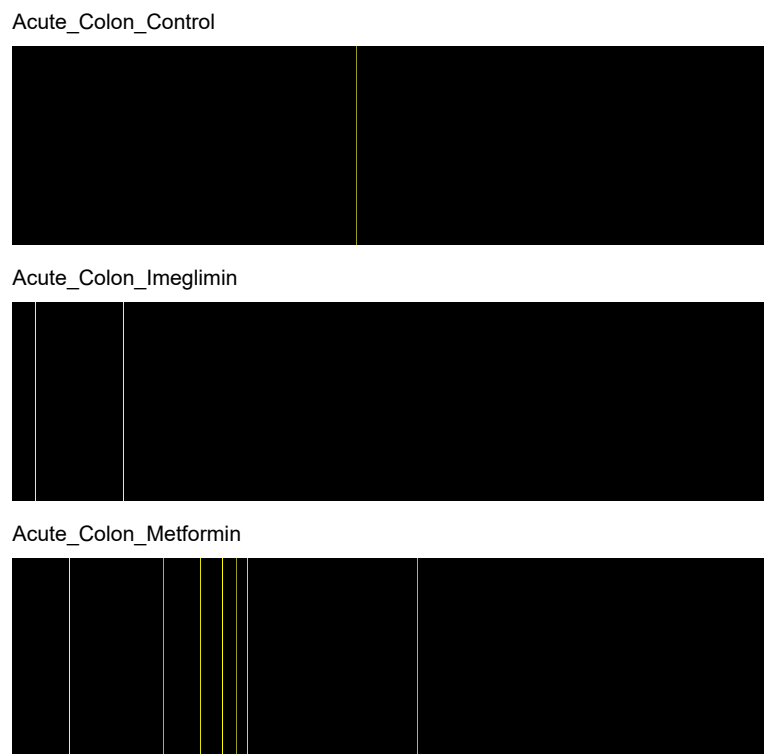


Figure S.10: Heatmaps for group Acute Colon Control, Acute Colon Imeglimin and Acute Colon Metformin.

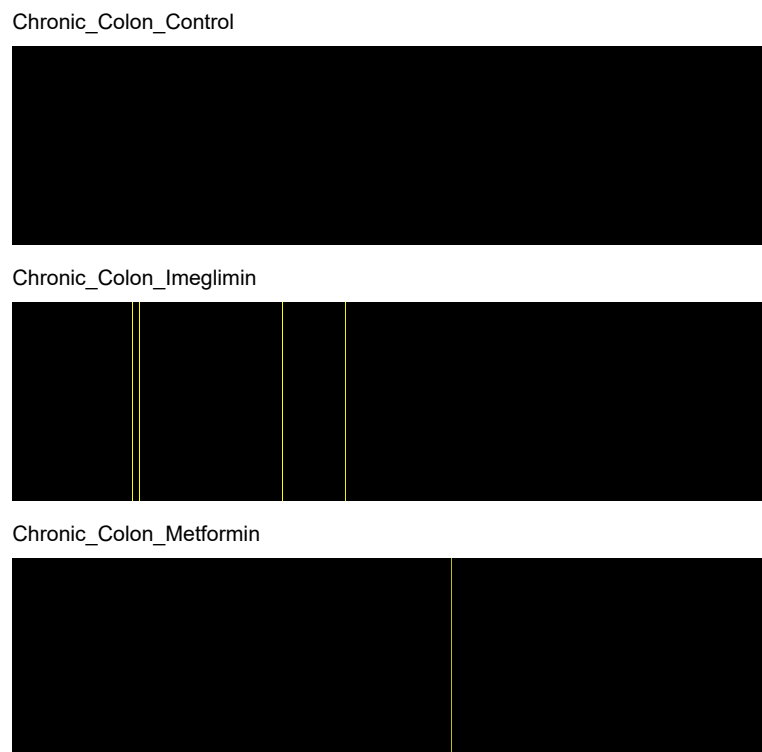


Figure S.11: Heatmaps for group Chronic Colon Control, Chronic Colon Imeglimin and Chronic Colon Metformin.

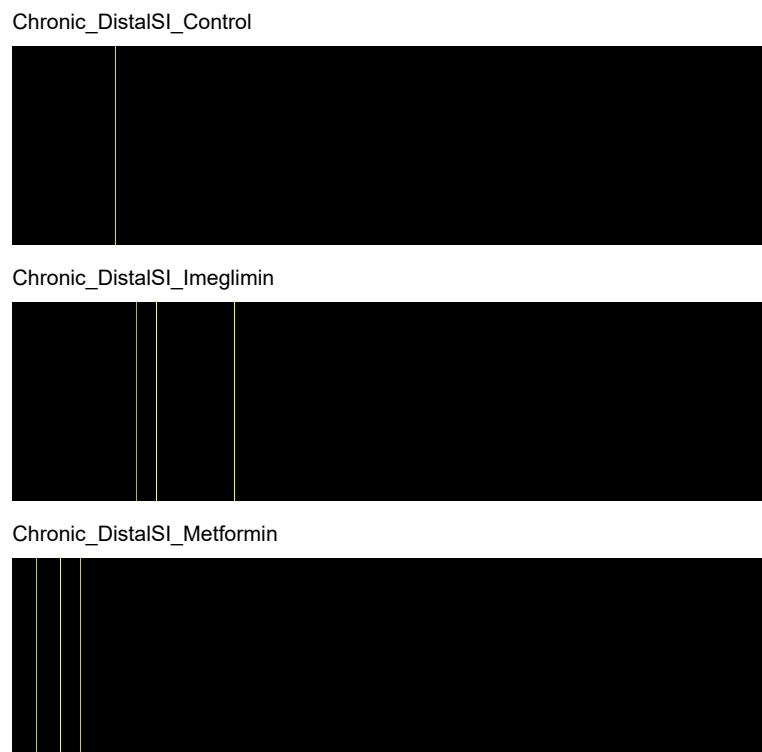


Figure S.12: Heatmaps for group Chronic DistalSI Control, Chronic DistalSI Imeglimin and Chronic DistalSI Metformin.

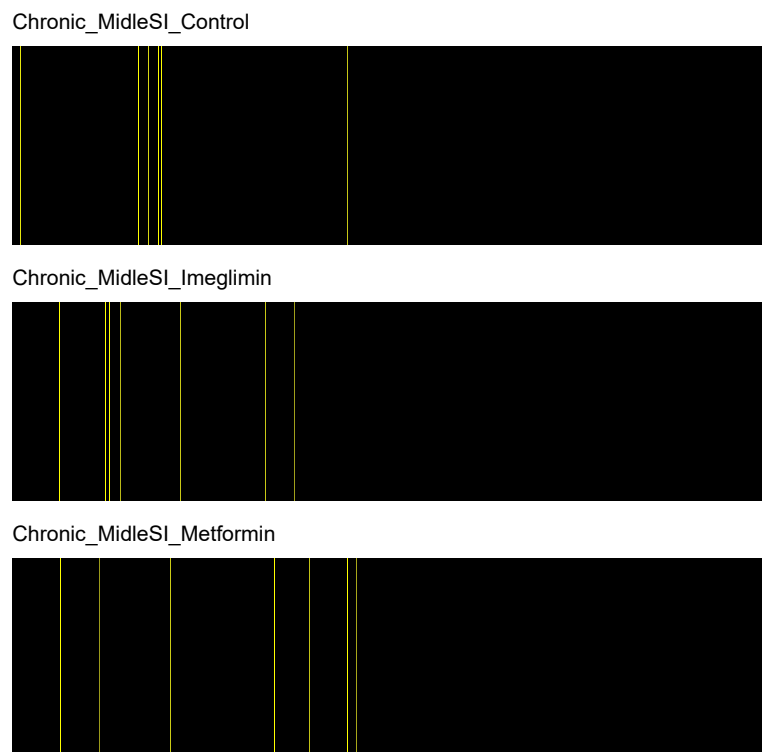


Figure S.13: Heatmaps for group Chronic MidleSI Control, Chronic MidleSI Imeglimin and Chronic MidleSI Metformin.

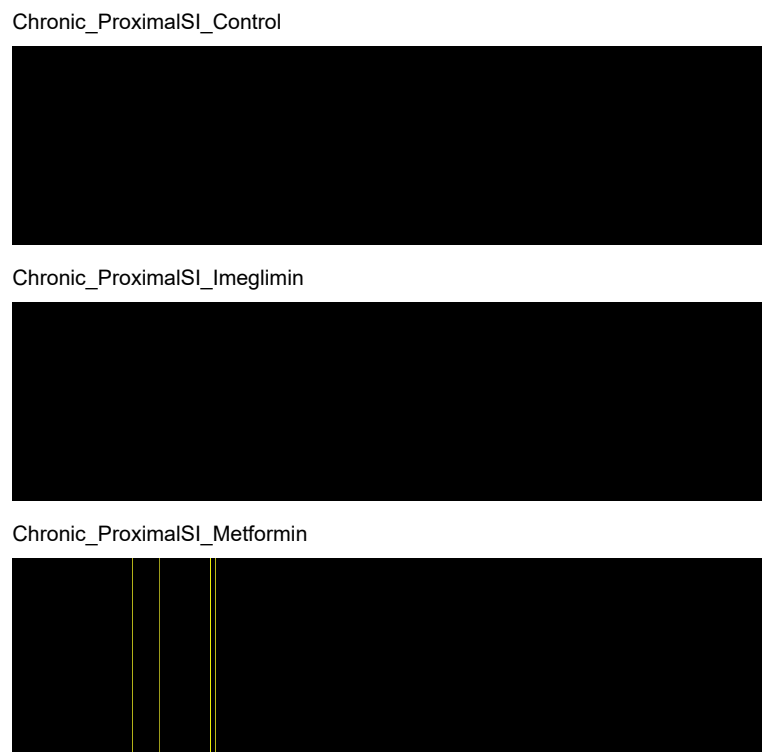


Figure S.14: Heatmaps for group Chronic ProximalSI Control, Chronic ProximalSI Imeglimin and Chronic ProximalSI Metformin.